

GENERAL SPECIFICATION G4.72

## Odour Control Systems

### CONTENTS

|          |                                                       |    |
|----------|-------------------------------------------------------|----|
| G4.72    | Odour Control Systems .....                           | 1  |
| G4.72.1  | General.....                                          | 1  |
| G4.72.2  | Standards and Reference Documents .....               | 1  |
| G4.72.3  | Definitions and Abbreviations .....                   | 3  |
| G4.72.4  | Additional Submittal Requirements.....                | 4  |
| G4.72.5  | Manufacturer’s Qualifications .....                   | 5  |
| G4.72.6  | System Guarantee.....                                 | 7  |
| G4.72.7  | Performance Guarantee .....                           | 7  |
| G4.72.8  | General Design Requirements - Overall.....            | 8  |
| G4.72.9  | Activated Carbon Scrubber Design Requirements.....    | 14 |
| G4.72.10 | Bio-Trickling Filter (BTF) Design Requirements .....  | 21 |
| G4.72.11 | Chemical Scrubber Design Requirements.....            | 27 |
| G4.72.12 | Biofilter Design Requirements .....                   | 34 |
| G4.72.13 | Mechanical Equipment Design Requirements.....         | 42 |
| G4.72.14 | Scrubber Vessel Design Requirements .....             | 57 |
| G4.72.15 | Product Delivery, Storage and Handling .....          | 64 |
| G4.72.16 | Specific Testing and Commissioning Requirements ..... | 65 |

## **G4.72 ODOUR CONTROL SYSTEMS**

### **G4.72.1 General**

- (a) This specification covers the general requirement of the performance, design and construction of odour treatment units for the neutralisation or absorption of odorous compounds by physical, chemical or biological action in used water (wastewater) application with all relevant statutory regulations and the latest edition of all relevant international, harmonised European and British Standards and Singapore Standards and Code of Practices. The materials of construction must be 100 % compatible for corrosion resistance to compounds typically associated with industrial effluent and sewerage foul air conditions inclusive of associated system reagents.
- (b) The specification covers the following type of treatment plant:
  - (i) Activated Carbon (AC) media scrubbers;
  - (ii) Bioscrubbers (i.e. Bio-Trickling Filter (BTF));
  - (iii) Wet Chemical Scrubbers; and
  - (iv) Biofilters.
- (c) This specification shall also comply with:
  - (i) The relevant Mechanical Particular Specification;
  - (ii) All associated General Specifications;
  - (iii) The specific technical data schedules accompanying the Particular Specifications (if applicable);
  - (iv) The relevant Electrical General and Particular Specifications; and
  - (v) The relevant Instrumentation and Control and Automation (ICA) General and Particular Specification.

### **G4.72.2 Standards and Reference Documents**

- (a) Equipment and components shall be in accordance with the latest version of applicable codes and regulations, including the following:
  - (i) Singapore Standards;
  - (ii) British Standards (BSI);
  - (iii) European Standards (EN);
  - (iv) International Standards Organisation (ISO) Standards;
  - (v) Statutory regulations and codes;
  - (vi) Occupation Health and Safety (OH&S) requirements; and
  - (vii) The Board's (PUB's) specifications.

- (b) This specification shall be read in conjunction with the following Standards, which are deemed to form a part of this specification. Reference to a Standard shall mean that Standard current as at the time of providing the works. All materials, equipment and workmanship shall comply with these Standards or other industry recognised standards in the absence of specified standards, unless expressly noted otherwise. Where requirements in the specifications are inconsistent or conflicting, the more stringent requirement shall take precedence.

| Standard       | Subject                                                                                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D6646-03  | Standard Test Method for Determination of the Accelerated Hydrogen Sulphide Breakthrough Capacity of Granular and Pelletized Activated Carbon.                              |
| BS EN 13725    | Air quality. Determination of odour concentration by dynamic olfactometry.                                                                                                  |
| BS EN 13923    | Filament-wound Fibreglass Reinforced Plastic (FRP) pressure vessels. Materials, design, manufacturing and testing.                                                          |
| BS 4994:1987   | Specification for design, construction, inspection, testing and erection of vessels and tanks in reinforced plastics.                                                       |
| BS EN 13121.1  | Glass-reinforced thermosetting plastics (GRP) tanks and vessels for use above ground. Raw materials. Specification conditions and acceptance conditions.                    |
| BS EN 13121.2  | GRP tanks and vessels for use above ground. Composite materials. Chemical Resistance.                                                                                       |
| BS EN 13121-3  | GRP tanks and vessels for use above ground. Design and workmanship.                                                                                                         |
| BS EN 13121.4  | GRP tanks and vessels for use above ground. Delivery installation and maintenance.                                                                                          |
| BS EN 14364    | Plastics piping systems for drainage and sewerage with or without pressure. (GRP) based on unsaturated polyester resin (UP). Specifications for pipes, fittings and joints. |
| BS EN 1796     | Plastics piping systems for water supply with or without pressure. GRP based on unsaturated polyester resin (UP).                                                           |
| BS EN ISO 5801 | Industrial fans - Performance testing using standardized airways.                                                                                                           |
| ISO 10780      | Stationary source emissions - Measurement of velocity and volume flow rate of gas streams in ducts.                                                                         |
| BS ISO 14164   | Stationary source emissions – Determination of the volume flow rate of gas streams in ducts – Automated method.                                                             |

| Standard     | Subject                                                                                                                                            |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| BS ISO 10396 | Stationary source emissions. Sampling for the automated determination of gas emission concentrations for permanently-installed monitoring systems. |

#### G4.72.3 Definitions and Abbreviations

- (a) For the purposes of this specification, the following list of abbreviations and definitions of units apply:

| Definitions / Abbreviations | Subject                                                                                                                                     |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| AC                          | Activated Carbon                                                                                                                            |
| AMCA                        | Air Movement and Control Association                                                                                                        |
| BTF                         | Bio trickling Filter                                                                                                                        |
| °C                          | Degrees Celsius                                                                                                                             |
| CO                          | Carbon Monoxide                                                                                                                             |
| dB(A)                       | Decibel, A scale                                                                                                                            |
| DMS                         | Dimethyl Sulphide                                                                                                                           |
| DMDS                        | Dimethyl Disulphide                                                                                                                         |
| DN                          | Nominal Diameter                                                                                                                            |
| EBRT                        | Empty Bed Residence Time                                                                                                                    |
| EPR                         | Ethyl Propylene Rubber                                                                                                                      |
| FRP                         | Fibreglass Reinforced Plastic                                                                                                               |
| GCMS                        | Gas Chromatography Mass Spectrometry                                                                                                        |
| HDPE                        | High Density Polyethylene                                                                                                                   |
| HDT                         | Heat Distortion Temperature                                                                                                                 |
| H <sub>2</sub> S            | Hydrogen Sulphide                                                                                                                           |
| IEC                         | International Electrotechnical Commission                                                                                                   |
| I/O                         | Input/Output                                                                                                                                |
| MCC                         | Motor Control Centre                                                                                                                        |
| Am <sup>3</sup> /hr         | Actual Cubic Metres per Hour, i.e. gas flow at actual gas temperature and pressure                                                          |
| Nm <sup>3</sup> /hr         | Normal cubic metres per hour, Air flowrate standardised to 20 degrees C, 101.35 kPa, and 0 % relative humidity, unless otherwise specified. |
| NDT                         | Non-Destructive Testing                                                                                                                     |

| Definitions / Abbreviations | Subject                                   |
|-----------------------------|-------------------------------------------|
| NEA                         | National Environment Agency               |
| OCF                         | Odour Control Facility                    |
| OCS                         | Odour Control System                      |
| OH&S                        | Occupational Health and Safety            |
| ou                          | Odour unit as defined in EN 13725         |
| P.E.                        | Professional Engineer                     |
| PLC                         | Programmable Logic Controller             |
| PUB                         | Public Utilities Board                    |
| PVC                         | Polyvinyl Chloride                        |
| P&ID                        | Process and Instrumentation Diagram       |
| ppm                         | Parts Per Million                         |
| ppm <sub>v</sub>            | Parts Per Million (volume basis)          |
| ppb                         | Parts Per Billion                         |
| ppb <sub>v</sub>            | Parts Per Billion (volume basis)          |
| Pa                          | Pascals                                   |
| kPa                         | Kilopascals                               |
| RH %                        | Relative Humidity Percentage              |
| RSC                         | Reduced Sulphur Compounds                 |
| RTD                         | Resistance Temperature Detector           |
| SAC                         | Singapore Accreditation Council           |
| SINGLAS                     | Singapore Laboratory Accreditation Scheme |
| S.O.                        | Superintendent Officer                    |
| SPT                         | Site Performance Testing                  |
| VOC                         | Volatile Organic Compound                 |
| WLL                         | Working Load Limit                        |

#### **G4.72.4 Additional Submittal Requirements**

- (a) The Contractor shall provide the following data and specifications for each equipment unit supplied, but not limited to, the information outlined in General Specifications on Information Management and the General Specification for Operation and Maintenance Manuals, and includes the following additional requirements:
  - (i) Details of any non-conformances relating to this technical specification;

- (ii) Manufacturer's catalogue information, operations and maintenance manuals, and relevant technical product brochures for proposed equipment, where available;
  - (iii) Clearly defined scope of supply and installation, including any assumptions;
  - (iv) Basic description of the odour control system offered;
  - (v) A list of priced and un-priced spare parts for both commissioning and 12-month operation. i.e. 12 month and 24 months spares recommendations;
  - (vi) Proposed media type and material, bed depths and retention time;
  - (vii) Any requirement for media re-generation (if applicable);
  - (viii) The potable water demand and plant water demand (flow and pressure requirement), along with any reject/blowdown water quantities and characteristics (if applicable);
  - (ix) The estimated energy consumption and electrical drive loadings list;
  - (x) Completed and final Technical Specification Tables as noted in the relevant Particular Specifications; and
  - (xi) Process verifications to demonstrate that any odour treatment units provided meets the required performance criteria.
- (b) With respect to the above, where appropriate, the drawings shall include the minimum clearance distance required for the removal of plant and equipment and its associated components (including media replacement) with the equipment in situ and termination points for the connection of external services.

**G4.72.5 Manufacturer's Qualifications**

- (a) The materials covered by these specifications are intended to be standard equipment of proven reliability and as manufactured by a reputable manufacturer having experience in the production of such equipment. The equipment furnished shall be designed, constructed, and installed in accordance with the best practices and methods and shall operate satisfactorily when installed as shown in the Drawings and operated per Manufacturer's recommendations.
- (b) The Manufacturer's Quality Assurance systems shall demonstrate traceability of materials back to the original equipment and material provider. Work procedures shall be provided by the Manufacturer which comply with the original equipment provider's equipment warranty conditions.
- (c) Unless otherwise specified, the manufacturer shall have manufactured at least THREE (3) comparable installations for the technology configuration offered, under comparable service conditions. Each comparable installation shall have been operating successfully and satisfied its associated process guarantees for a minimum of FIVE (5) years.

- (i) The Manufacturer shall provide a list of at least THREE (3) air treatment installations utilizing the offered technology configuration associated with the removal of Hydrogen Sulphide ( $H_2S$ ) and other typical organic used water odours for which they have been responsible. This schedule, along with supporting documentation (e.g.  $H_2S$ , Volatile Organic Compounds (VOC), and independent olfactometry test results) must be included with the submitted offer. Proof of integrity of the FRP vessel jointing system, with references, shall be provided;
  - (ii) The list shall include contact names, telephone numbers, system capacity (flow rates in  $Nm^3/hr$ , removal efficiency of system in terms of odour and  $H_2S$  (OU and  $ppm_v$  in/out, and percentage removal), media life, type of media used, and length of operation for each installation;
  - (iii) The referenced installations must be for the odour control technology configuration offered by the Manufacturer;
  - (iv) The referenced installations should have process units of a comparable size and treatment load; and
  - (v) The list shall verify to the S.O. that the specified manufacturer's qualifications requirements are met.
- (d) Products
  - (i) This section covers the various possible technologies for odour control equipment which may be supplied by the Contractor, which includes:
    - 1) AC media scrubbers;
    - 2) Bio-Trickling Filters;
    - 3) Wet Chemical Scrubbers
    - 4) Biofilters
  - (ii) Specific odour control technology requirements (if any) will be outlined in the Particular Specifications for each site;
  - (iii) The Contractor shall only offer those proven technologies as listed above with references as its primary offer. While a new technology may be available and may be considered by the S.O., primary offers which include unproven technologies will be rejected during the review process; and
  - (iv) Required instrumentation and controls for the Odour Control System (OCS) shall comply with General Specification G6 ICA. The minimum level of instrumentation required for the OCS is shown indicatively on the preliminary process and instrumentation diagrams (P&ID) provided with this Technical Specification. The Contractor shall supply a P&ID showing all equipment and instruments which will be provided as part of this tender for the S.O.'s acceptance. All equipment and instruments to be supplied as part of the OCS shall be shown on the P&IDs supplied. The Contractor shall also supply their intended operating control philosophy for the S.O.'s approval and for subsequent use by the S.O. in programming the plant PLC which shall control the OCS.

#### **G4.72.6 System Guarantee**

- (a) The Contractor shall warrant that:
  - (i) All equipment provided by the Contractor as part of the OCS package shall be in accordance with this specification and shall perform as specified in the Technical Specification tables in the Particular Specifications;
  - (ii) The OCS shall operate automatically at an unattended facility, monitored remotely via The Board's Monitoring & Control System. All equipment supplied by the Contractor shall be suitable for unattended operation;
  - (iii) Based on the anticipated air flows through the OCS and the ambient conditions of the installation sites, all equipment items forming part of the OCS shall retain their structural and functional integrity for a period of twenty (20) years from the date of practical completion; and
  - (iv) The Contractor shall repair or provide replacements for any defective components under the warranty provided that the defect is not the result of misuse by the Board. Replacements for any defective parts shall be provided by the Contractor to restore equipment to its "as commissioned" working condition.

#### **G4.72.7 Performance Guarantee**

- (a) The Contractor shall guarantee that all equipment forming part of the OCS proposed shall achieve the following performance:
  - (i) The proposed odour control technology presented by the Contractor shall be in accordance with the requirements as outlined in the Particular Specifications.
  - (ii) The H<sub>2</sub>S, odour, and other contaminant (VOC, RSC) removal performance shall be in accordance with the specified removal efficiencies referenced in the Performance Guarantees outlined in the Particular Specifications, as measured between the OCS gas inlet and the discharge point (i.e. exhaust stack or Biofilter outlet).
  - (iii) The maximum outlet H<sub>2</sub>S and odour concentration at the exhaust location (i.e. exhaust stack or at the surface of Biofilters if supplied) must not exceed the guaranteed concentrations as specified in the Performance Guarantees outlined in the Particular Specifications under all maximum design operating concentrations of H<sub>2</sub>S and odour at the inlet to the OCS.
  - (iv) If stipulated in the Particular Specifications, the maximum H<sub>2</sub>S concentrations at the site boundary as demonstrated by air dispersion modelling using an appropriate model (such as 3D CALPUFF) by the Contractor must not exceed the guaranteed concentrations as specified in the Particular Specifications Performance Guarantees under all maximum design operating concentrations of H<sub>2</sub>S and odour at the inlet to the OCS.
  - (v) The maximum pressure differential across each OCS component shall not exceed that specified in the Particular Specifications for all design conditions throughout the life of the installation, when measured between the OCS gas inlet and outlet.
  - (vi) Media Life



- 1) Unless otherwise specified in the Particular Specifications, the media life for AC systems shall be no less than twelve (12) months for each AC vessel, to H<sub>2</sub>S and/or odour breakthrough, defined as when the outlet H<sub>2</sub>S and/or Odour concentrations exceeds the guaranteed concentration limits specified in the Particular Specifications Performance Guarantees and Technical Data Schedules; and
  - 2) Unless otherwise specified in the Particular Specifications, the media life period for Bio-Trickling Filter systems shall be no less than ten (10) years for each BTF vessel, to media breakdown and deterioration.
  - 3) Unless otherwise specified in the Particular Specifications, the media life period for engineered media Biofilter systems shall be no less than ten (10) years, to media breakdown and deterioration.
- (vii) Process performance testing comprising of H<sub>2</sub>S and odour removal performance testing shall be the means of proving the performance criteria outlined in the Schedule of Performance Guarantees stipulated in the Particular Specifications has been met following installation and commissioning of the OCS. H<sub>2</sub>S and odour removal performance testing must be undertaken by an accredited laboratory for the results to be incorporated into the performance testing results.
- (viii) The details and timing of the proposed performance testing can be found in G4.72.16(f).

#### **G4.72.8 General Design Requirements - Overall**

(a) General

- (i) Details of the plant and equipment requirements shall refer to the Particular Specifications.
- (ii) Unless otherwise specified in the Particular Specifications, the odour treatment system will normally operate continuously and in a remote unmanned site, i.e. 24 hours per day, 365 days per year, reliably.
- (iii) Unless otherwise specified in the Particular Specifications, adequate equipment redundancy shall be provided by the Contractor to allow a minimum of ONE (1) duty mechanical equipment item (e.g. odorous air fans, irrigation pumps, etc.) and/or process vessel (e.g. AC scrubber vessel, BTF vessel, Chemical Scrubber, Biofilter cell, etc.) to be taken offline for maintenance and/or replacement without causing process upsets or critical failures.
- (iv) The plant layout drawing is provided for the information of the Contractor. It shows general dimensions and locations of features that the Contractor shall incorporate into the final design. Any changes to these features that the Contractor may find necessary in the design or detail work must be approved by the S.O..
- (v) Preliminary P&IDs are provided for the information of the Contractor. It shows the expected equipment which shall be supplied by the Contractor and which will need to be incorporated into the final design.

(b) Minimum Equipment Items

- (i) The odour treatment units shall include as a minimum, but is not limited to, the following equipment items:
  - 1) Odour covers to contain the odour air at its sources (Refer to General Specification on Odour Containment Covers - G4.73).
  - 2) An inlet manifold and associated ductwork to receive odorous air from the specified extraction points and stipulated ventilated sources which direct it to the odour treatment units for treatment.
  - 3) The odour treatment units along with its associated ancillaries, including but not limited to: containers, tanks, access stairways/ladders, walkways and hand railings, pumps, fans, nozzles, filters, cablings, trunkings, control and instrumentation, pipework, flow control dampers, isolation dampers, valves, air distribution plenum, fittings and its supports etc. Access to the top of BTF, AC vessels, Chemical Scrubber Vessels, and Biofilter cells will be required, and adequate access staircases, platforms, and handrails must be provided for this purpose. Panels of FRP grating of adequate lengths shall be provided at the vicinity of the Biofilter cells to allow personnel access onto the Biofilter media without directly stepping on the media. Adequate access must also be provided to instrumentation, pumps, fans, control valves, and isolation dampers etc. for calibration and maintenance purposes.
  - 4) If the odour treatment unit incorporates an Exhaust Stack, an outlet manifold and associated ductwork shall be provided to receive the treated air from the OCS and direct it to the Exhaust Stack for discharge.
  - 5) Extraction fans to extract air from the specified extraction points and convey it through the various odour treatment equipment.
  - 6) Associated electrical, instrumentation, and final control elements necessary to allow automatic operation of the odour control units. This includes analyser panels and associated monitoring facilities and equipment.
- (ii) Any AC vessels proposed shall consist of a single pass, vertical system, and may consist of single or dual beds installed in each vessel and supplied with all necessary internals to provide a complete functional system.
- (iii) Adequate condensate and spent liquor drainage facilities shall be provided for all process equipment (BTF, AC, and Chemical Scrubber vessels, also Biofilters, Fans and Pre-Filters), and at strategic points along odour ductwork. Drainage points shall incorporate a water sampling point and water trap for containment of process air. Specific design requirements drainage points are listed below and common across all drainage lines:
  - 1) The drainage point (BTF vessel, AC vessel, Chemical Scrubber vessel, Biofilter cell, Pre-Filters, or odour ductwork) shall be fitted with a suitable sized drain line not less than DN25 with a “U” or “S” bend water trap to drain leachate and prevent air release;

- 2) The “U” or “S” bend shall be made of clear plastic so that visual confirmation of the seal fluid can be undertaken. The drain line must contain a tapping on the “U” or “S” bend for inspection and wash-down if required; and
  - 3) All drain points shall be rated for a minimum of 1.5 times the pressure at the point of installation (ducting, stack, BTF, AC, Chemical Scrubber vessel, or Biofilter cells) and be made of materials resistant to corrosion by the condensate, which may be highly acidic.
- (iv) A bund shall be provided around the OCF reactor vessels to provide containment in the unlikely event of spillages. Depending on the location of the OCF, surface water may be collected in the bund sump area. The following requirements apply to the bund sump area:
- 1) Surface water collected in any of the bund sumps shall be directed to the nearest Surface Water Drainage Point for direct discharge
  - 2) The Contractor shall configure the OCF Bund Sumps such that where there are spillages of process fluid in the bund area, discharge to the nearest surface drain will be stopped, and the process fluid drained to the nearest plant drain pump station for return to the nominated head of works.
- (c) Materials of Construction
- (i) The materials of construction must be 100% compatible for corrosion resistance to compounds typically associated with sewerage and/or industrial effluent odorous air conditions (as applicable) and climatic conditions at Singapore. These compounds and expected climatic conditions are noted in the Particular Specifications for each site.
    - 1) Unless otherwise specified or agreed with the S.O., the only acceptable materials of construction for odour ducting or process units comprise of Vinyl Ester Fiberglass Reinforced Plastic (FRP).
    - 2) Where concrete is used for the Biofilter base and walls, adequate corrosion protection in the form of a corrosion resistant barrier in the form of protective coatings or plastic liners shall be provided over all exposed concrete structures as outlined in the Particular Specifications. The concrete furnished for the Biofilter vessels shall satisfy the requirements outlined in General Specifications G3.20 Concrete
  - (ii) The materials of construction to be used on all components of the OCS are shown in Table G4.72-1 below unless otherwise specified in the Particular Specifications. Any deviations to the materials specified shall be submitted to S.O. for prior approval.

**Table G4.72-1 – OCS Table of Materials of Construction**

| Item  | Component                      | Acceptable Material        |
|-------|--------------------------------|----------------------------|
| Tanks | Vessels                        | FRP with vinyl ester resin |
|       | Vessel external nuts and bolts | 316 Stainless Steel        |

| Item                                                             | Component                                                      | Acceptable Material                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Odour Scrubbers (BTF, AC, and Chemical Scrubber Vessels)         | Vessels                                                        | FRP with vinyl ester resin.<br>See Section G4.72.14(f) for detailed requirements<br><br>Internal protective coating resin system in accordance with BS 4994: 1987                                                                                                                                                                                                                                                                              |
|                                                                  | Internal support beams (or equivalent)                         | FRP with vinyl ester resin.<br>See Section G4.72.14(f) for detailed requirements                                                                                                                                                                                                                                                                                                                                                               |
|                                                                  | Internal media support plates/carriage trays/baskets           | FRP with vinyl ester resin.<br>See Section G4.72.14(f) for detailed requirements<br><br>FRP grid or other corrosion resistant material. Sufficient to hold required media and entrained liquid.<br><br>Polypropylene screen material (i.e. AC). Opening size shall not allow passage of media or supported biomass (where applicable).<br><br>Structurally supported to limit the mid-span deflection to (L/125) where (L) is the span length. |
|                                                                  | Turning vanes                                                  | FRP with vinyl ester resin.<br>See Section G4.72.14(f) for detailed requirements                                                                                                                                                                                                                                                                                                                                                               |
|                                                                  | Distribution pipework                                          | FRP with vinyl ester resin.<br>See Section G4.72.14(f) for detailed requirements                                                                                                                                                                                                                                                                                                                                                               |
|                                                                  | External nuts and bolts (including anchor bolts)               | 316 Stainless Steel                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                  | Irrigation Spray Nozzles (Incl. Firefighting Sprinkler System) | Non-clogging type, PVC or polypropylene.                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                                  | Gaskets and seals                                              | EPDM                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Access Ports, Hatches, and Manways on Scrubber Vessels and Ducts | Fixings and fastenings                                         | FRP or 316 Stainless Steel                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                  | Hatches/Manways                                                | FRP with vinyl ester resin access openings shall be provided to allow inspection of the mist eliminator, sump, access for media replacement and irrigation spray nozzles on each vessel (refer G4.72.13(g)).<br><br>Clear polycarbonate covers with backing rings for observation as outlined in G4.72.13(g).                                                                                                                                  |

TUAS WATER RECLAMATION PLANT  
CONTRACT 4B – DIGESTION PRE-TREATMENT AND  
SIDESTREAM NITROGEN REMOVAL FACILITIES

VOLUME 2

| Item                         | Component                                | Acceptable Material                                                                                                                                                                                                                                    |
|------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              |                                          | Bolts shall be Type 316 SS and installed using nickel anti-seize                                                                                                                                                                                       |
|                              | Manway cover lifting lugs                | 316 Stainless Steel                                                                                                                                                                                                                                    |
|                              | Seals                                    | EPDM or Silicone Rubber                                                                                                                                                                                                                                |
|                              | Bolts                                    | 316 Stainless Steel installed using nickel anti seize                                                                                                                                                                                                  |
| BTF Irrigation Water Cabinet | Panel                                    | 316 Stainless Steel                                                                                                                                                                                                                                    |
| Pre-Filter                   | Housing                                  | FRP with vinyl ester resin or 316 Stainless Steel                                                                                                                                                                                                      |
| Fan                          | Impeller                                 | FRP with vinyl ester resin See section G4.72.13(b) for detailed requirements                                                                                                                                                                           |
|                              | Shaft                                    | Steel polished carbon steel or polished 316 SS. The portion of the shaft exposed to the airstream will be FRP encapsulated. The neoprene shaft seal in the fan housing will provide a seal against fumes migrating in or out of the shaft penetration. |
|                              | Housing                                  | FRP with vinyl ester resin                                                                                                                                                                                                                             |
|                              | Fan Support                              | Manufactured entirely out of epoxy coated mild steel                                                                                                                                                                                                   |
|                              | Inlet and outlets                        | FRP flanged inlets and outlets, with 316 Stainless Steel draw bands for attachment of flexible connections.                                                                                                                                            |
|                              | Fasteners and nameplates                 | 316 Stainless Steel                                                                                                                                                                                                                                    |
|                              |                                          |                                                                                                                                                                                                                                                        |
| Acoustic enclosure           | Outer skin                               | Thick FRP with vinyl ester resin or 316 Stainless Steel                                                                                                                                                                                                |
|                              | Inner skin                               | Perforated 316 Stainless Steel                                                                                                                                                                                                                         |
|                              | Filling                                  | High density mineral wool                                                                                                                                                                                                                              |
| Ductwork                     | Ducting                                  | FRP with vinyl ester resin See section G4.72.13(a) for detailed requirements                                                                                                                                                                           |
|                              | Clamps and brackets etc.                 | 316 Stainless Steel                                                                                                                                                                                                                                    |
|                              | Gaskets, seals, and flexible connectors. | EPDM                                                                                                                                                                                                                                                   |
| Dampers                      | Damper                                   | FRP with vinyl ester resin                                                                                                                                                                                                                             |
|                              | Spindle                                  | 316 Stainless Steel<br>See section G4.72.13(e) for detailed requirements                                                                                                                                                                               |

| Item                              | Component                             | Acceptable Material                                                              |
|-----------------------------------|---------------------------------------|----------------------------------------------------------------------------------|
| Exhaust Stack                     | Stack                                 | FRP with vinyl ester resin.<br>See Section G4.72.13(c) for detailed requirements |
|                                   | Bolts                                 | 316 Stainless Steel                                                              |
| Access platform                   | -                                     | See General Access Specification                                                 |
| Condensate / leachate drain lines | All condensate / leachate drain lines | cPVC                                                                             |

(d) Noise Levels

- (i) Unless otherwise specified in the Particular Specifications, noise levels from the combined components comprising the OCS (i.e. fans together with ductwork and other associated equipment within the immediate vicinity) shall satisfy the relevant OH&S regulatory requirements for safe operation.

(e) Air Balancing

(i) General

- 1) Air handling systems, ductwork, and headers shall be air balanced to give flow rates within 5% of designated air quantities, subject to the following:
  - Air shall be distributed to minimise pressure loss per unit length (Pa/m) and duct velocities; and
  - Air resistance at fans shall be minimised, and the fans shall be adjusted to run at lowest fan speeds and power consumption.
- 2) Balancing points shall be provided in ducts in sufficient number to facilitate the proper testing and commissioning of the air collection system, with a minimum of one balancing point for each branch duct and each main duct.
- 3) All balancing points shall be located in readily accessible positions in straight ducts, at least seven 'hydraulic diameters' downstream from any fitting, bend or air control device. Where the required distance (seven 'hydraulic diameters') cannot be achieved, the Contractor shall design, supply and install straightening vanes or diffusing grid to achieve stable flow conditions at the balancing point. All balancing points shall be easily accessible for measurement and provided with a minimum of two (2) tappings at ninety degrees (90°) apart of adequate size for insertion and installation of a handheld flow measurement device.

(ii) Procedure

- 1) Air balancing shall commence after the entire odour containment covers and OCS installation have been completed;
- 2) Where there are inter-connecting ducting header systems, all inter-connected systems shall be operated concurrently; and

- 3) Final position of flow balancing dampers shall be marked upon completion of balancing and locked into position.

(iii) Reporting

- 1) The following shall be included on the air balance data sheets:
  - Static pressure differentials,
  - Air quantities through each sub-system or branch ducting after balancing, and
  - Fan capacity and fan speed.
- 2) The final operating point shall be shown on the fan characteristic curve.

**G4.72.9 Activated Carbon Scrubber Design Requirements**

(a) General

- (i) This section covers odour control equipment where odorous air is passed through AC based media and odorous compounds are adsorbed or neutralised by the media. The performance may be enhanced using adsorption AC impregnated with a chemical agent or catalyst.
- (ii) When the AC beds are the primary treatment process adopted and VOC removal is required, inlet heaters/dehumidifiers shall be installed upstream of the AC scrubber. The Manufacturer shall specify the required dehumidifier duty operating range to achieve the target inlet air relative humidity required for acceptable VOC removal through the AC scrubber.
- (iii) The complete AC odour treatment system as proposed by the Contractor shall fit within the designated areas as outlined in the specifications.
- (iv) The AC systems shall be a vertical bed, single pass system, (multiple beds installed in each vessel acceptable), and with all necessary internals, fixtures, fittings, and support structures to provide a complete functional system.
- (v) The vessel dimensions (length x width and/or diameter) and media volume shall be calculated by the Contractor and properly designed to achieve the specified Performance Guarantees stipulated in the Particular Specifications.
- (vi) All equipment shall be designed to allow safe service access for maintenance and removal, and for replacement of assemblies and component parts, without disturbance of other items at the OCS. The following minimum distances between adjacent AC vessels shall be satisfied to allow adequate access for maintenance and AC media replacement:
  - 1) All AC vessels shall be installed adjacent to a dedicated service road or grassed area capable of passing standard sized vehicles and trucks;
  - 2) Dual-bed and single-bed AC vessels mounted vertically shall be provided with at least 1.5 m between adjacent AC vessels for access;

- 3) AC reactors shall be designed such that extra depth of AC media or addition of AC vessels may be added to meet performance requirements, should the initial odour performance testing results show failure to meet the guaranteed contaminant removal targets or the guaranteed emission limits. The addition of extra AC media depth or AC vessels to meet performance requirements will be at the Contractor's cost; and
  - 4) The AC reactors shall be installed and mounted such that it is not necessary to disassemble the vessel for access to remove and replace the media in each AC bed.
- (vii) The AC systems shall incorporate the following ancillary systems as minimum:
- 1) Inlet pre-filters; and
  - 2) Inlet heaters/dehumidifiers (specific to Contractor's proposed AC blend and AC system operating conditions);
- (viii) For AC based systems, the following interlocks and alarms must be built into the Odour Treatment Plant Control System and local operator interface, in addition to the standard alarms required for operation of the entire OCS:
- 1) For each AC bed:
    - High AC bed temperature alarm and High AC Carbon Monoxide (CO) sensor alarm, to be interlocked with operation of the AC bed emergency fire quench system and trigger its operation;
    - High-high AC bed temperature alarm, to be interlocked with operation of the AC bed emergency fire quench system and trigger its operation, irrespective of the CO sensor reading; and
    - High-High AC bed differential pressure alarm, to raise an alarm when an abnormal condition is detected, such as plugging of the AC media bed suggesting that media replacement may be required;
- (b) AC Media Requirements
- (i) The type and composition of the AC media required / provided shall be subject to the Contractor's specifications as specified in the Particular Specifications for the project site.
  - (ii) The Contractor shall be responsible for the appropriate carbon selection for the duty airflow and contaminants specified. The Contractor shall provide justification for the carbon selection including calculations to demonstrate that the minimum requirements will be achieved and supported by case studies or references which must be submitted to validate claims.
  - (iii) The AC media shall be of granular or pelletized form, and its composition shall match those specified in the Particular Specifications, with no more than 5 % variation in weight for each component making up the AC media.
  - (iv) The AC media depth shall provide a minimum ventilation air contact time within the AC media bed as specified in the Particular Specifications, with no more than 5 % compaction or shrinkage of media within the bed. The maximum acceptable media depth for AC media is 1,200 mm.



- (v) The depth of media shall be such that the minimum contact time for the airstream in the media shall satisfy the requirements outlined in Particular Specifications for each site, at the maximum design airflow, without exceeding the manufacturer recommended gas velocity through the AC media face. The AC media shall be evenly distributed in the bed so that no bypassing or short circuiting of odorous air occurs.
- (vi) The Contractor shall ensure that all AC media beds receive equal airflow and that no short circuiting occurs. Where multiple AC media beds or multiple AC cells are proposed, the contractor shall provide calculations and supporting documentation that demonstrate how equal airflow distribution will be achieved and how short circuiting will be eliminated or minimised.
- (vii) AC media adsorption H<sub>2</sub>S efficiencies (to odour or H<sub>2</sub>S break through, not saturation) based on the proposed media blend shall be supplied by the Contractor for reference. Recent extensive testing by the Australian Research Council (ARC) Sewer Corrosion and Odour Research (SCORE) project has identified that the typical H<sub>2</sub>S adsorption capacity to H<sub>2</sub>S breakthrough is much less than and can be as low as half of typical literature claims. Literature claims of AC media H<sub>2</sub>S adsorption are typically based on ASTM D6646-03 or equivalent, which quantifies the total H<sub>2</sub>S adsorption capacity of the media to saturation. This is impractical for field operation of OCSs as H<sub>2</sub>S and odour nuisance will be caused by H<sub>2</sub>S breakthrough, and will result in the need for AC media changeover. The Contractor shall adopt AC media efficiencies outlined in the table below in their design of proposed AC systems for H<sub>2</sub>S removal. The Contractor shall supply the adopted AC media efficiency in the design of proposed AC systems for VOC removal. The S.O. reserves the right to test AC media efficiency to breakthrough via adsorption column testing on random batches to validate efficiency claims.

| Carbon Type                                                                                | Adopted Values for H <sub>2</sub> S Adsorption Capacity (g/cm <sup>3</sup> media) |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Virgin Activated Carbon                                                                    | 0.014                                                                             |
| Chemically Impregnated Activated Carbon or Alumina (Hydroxide or Permanganate impregnated) | 0.063                                                                             |
| Catalytic Activated Carbon                                                                 | 0.072                                                                             |

- (viii) The Contractor is to ensure that each lot of AC media delivered to the project site for installation into the AC vessels complies with the requirements outlined in the Particular Specifications and is accompanied by an analysis sheet of the measured characteristics of the carbon in that lot.
- (ix) The Contractor shall provide information including details and price of the AC media intended to be supplied in the tender documents. The S.O. may agree to proceed with this intended carbon, however the S.O. reserves the right to request the Contractor to amend the specified carbon blend if deemed unsuitable for the contaminants in the odorous air requiring treatment.

- (x) The S.O. also reserves the right to source the proposed AC media blend independently from the Contractor, and not use the media offered by the Contractor. Should the S.O. agree to use the Contractor's carbon, it shall comply with the following requirements:
  - 1) Details of the proposed AC media blend, and its performance characteristics, particularly moisture ingress shall be provided by the Contractor. Peat, coconut shell or high-grade coal are the preferred base materials for carbon manufacture;
  - 2) The AC media shall also be required to absorb contaminants in addition to H<sub>2</sub>S and odour as specified in the Particular Specifications, that are present in the incoming air stream requiring treatment; and
  - 3) Unless otherwise specified in the Particular Specifications, a minimum of 50 % virgin activated carbon (or similarly suitable AC media) shall be incorporated into the AC media blend, for removal of Reduced Sulphur Compounds (RSCs) and VOCs which are typically resistant to adsorption onto impregnated activated carbon or catalytic activated carbon.
- (xi) Breakthrough and Media Replacement
  - 1) Breakthrough of H<sub>2</sub>S and odour is defined to occur when the H<sub>2</sub>S and odour reduction criteria in the Performance Guarantees stipulated in the Particular Specifications are exceeded, and consequently the AC media bed requires replacement;
  - 2) Bed life is defined as the length of time between replacement of AC media. Unless otherwise specified in the Particular Specifications, the Contractor shall guarantee the AC media bed life shall exceed one (1) year to break through of H<sub>2</sub>S and odour when operated continuously under the stated design inlet air conditions as outlined in the Particular Specification; and
  - 3) Unless otherwise specified in the Particular Specifications, the activated carbon shall have an odour removal efficiency of not less than 99 % and meet discharge H<sub>2</sub>S and odorous gas concentration limits as stipulated in the Schedule of Performance Guarantees in the Particular Specifications.
- (c) De-humidifiers/Heaters
  - (i) De-humidifiers/Heaters are typically required for odour treatment units utilising AC only. However, the requirement for a dehumidifier will be dependent upon the type of odorous air requiring treatment and AC media type proposed by the Contractor, given the different tolerances of differing AC media towards relative humidity (RH %).
  - (ii) Where VOC is the main constituent requiring treatment, de-humidifiers are typically required to maintain the relative humidity below the maximum tolerance for VOC removal from the incoming air.
  - (iii) The Contractor's submission shall indicate:
    - 1) Whether a dehumidifier/heater will be required for the specified OCF, with characteristics as written in this section; or

- 2) The reasoning behind why a dehumidifier/heater is not necessary for the specified OCF, with reference to the type of contaminants in the air to be treated and AC media blend proposed.
- (iv) Supplied dehumidifiers/heaters shall have the following characteristics:
  - 1) The dehumidifiers/heaters shall be comprised of electric heater element(s) in a suitable housing integrated with the supplied pre-filters;
  - 2) Dehumidifiers/heaters shall reduce the humidity by a minimum of 15 % RH and maintain relative humidity at the optimum RH required for removal of the specified contaminant at the design average air flowrate, and preferably less than 60 % RH;
  - 3) Be fitted with a thermostatically controlled heater element with a temperature indicator transmitter on the discharge stream and an integral over temperature protection;
  - 4) Low and high temperature alarms shall be transmitted to the local control panel of the plant at which it is installed, along with associated interlocks; and
  - 5) Be contained within an insulated self-contained sealed housing with lockable access hatches that are easily opened and closed for inspection and maintenance. The housing shall also have flanged inlet and outlet connections to ductwork and be insulated to minimise heat loss to the surroundings.
- (v) The dehumidifiers/heaters shall be suitably protected in the following manner:
  - 1) Electrically to prevent the housing becoming “live” in case of corrosion of the electric heating elements as well as protecting the elements from over-heating;
  - 2) Mechanically for OH&S considerations, to protect against personnel being exposed to hot surfaces; and
  - 3) The dehumidifiers/heaters shall be fitted with clearly labelled lockable electrical isolation switches mounted on an adjacent wall or purpose supplied pedestal such that the heater elements can be independently isolated electrically for maintenance or repair purposes.
- (vi) If the dehumidifier/heater is not required, the Contractor shall allow space within the OCF and pre-filter housing to allow a future connection and installation of a dehumidifier/heater within the pre-filter housing upstream of the AC vessel.
- (vii) Contractors are to provide details of the dehumidifier/heater and housing in the tender submissions for S.O. approval.
- (d) Fire Fighting Sprinkler System
  - (i) Where AC systems are provided for treatment of odour with high inlet H<sub>2</sub>S loads (greater than 10 ppm), or where hydroxide impregnated AC media is proposed, AC media beds must be fitted with a carbon bed overheat water quench facility connected to a continually present water supply from fire service water or equivalent quality site service water.

- (ii) The water supply line shall be fitted with a flow switch, which, when the quench system is activated, will send a signal to the local control panel and from there to The Board's remote monitoring system.
  - (iii) To ensure that the quench system is not activated due to the activated carbon unit being exposed to UV radiation, the activated carbon unit shall be fitted with a shade structure if required. The quench system will also be interlocked with the carbon monoxide (CO) sensors installed on the AC vessel common exhaust on readings of elevated temperature, to prevent the quench system being activated due to temperature rises caused by ambient weather conditions. A high-high temperature alarm will also automatically trigger operation of the fire quench sprays, irrespective of the reading from the CO sensor.
  - (iv) For the Fire Fighting sprinkler system where specified, it shall form part of the AC system package and shall consist of one or more spray nozzles dedicated to each AC bed. The system shall be designed to ensure that the entire surface area of each AC bed can be adequately reached by the firefighting sprinklers to quench any local fires which occur.
  - (v) Each firefighting spray bar and nozzles shall be arranged so that they can be removed from the AC vessel without needing to enter the tower or needing to remove any ductwork and shall be easily accessible from the adjacent access platform.
  - (vi) Nozzles shall be non-clogging type and constructed as per materials indicated in this Specification unless otherwise stated in the Particular Specifications.
- (e) Specific Design Requirements
- (i) The treatment unit shall be designed so that the volume of condensate accumulating within it is minimised. Facilities shall be provided with water traps for the collection and drainage of condensate to the point where it will prevent escape of odorous air.
  - (ii) The volume and arrangement of the media shall be such that the rates of chemical and physical reaction between the media and odorants within each treatment state are optimised to achieve the specified performance.
  - (iii) The AC vessel design shall employ methods to prevent caking, solidifying, and blooming of the installed AC media.
  - (iv) The configuration of the media bed shall be submitted with the Contractor's response to this tender. Carriage trays/baskets where provided, shall satisfy the following requirements:
    - 1) Be effectively seam sealed to their support structure; and
    - 2) Trays/baskets shall be designed so that they can be removed and replaced without damaging the treatment unit housing unit and without prolonged disruption to the odour treatment process.
  - (v) The installed AC media shall have sufficient compressive strength to resist all operational loads without more than 5 % deformation or bed settlement over its entire design life.

- (vi) Vessel Sample Ports
  - 1) For AC vessels, each AC scrubber vessel and/or AC media bed shall be fitted with the following additional sampling ports to collect air samples within each AC media bed, positioned at different bed depths (3 of), in addition to the inlet plenum and outlet plenum. Each port shall take the form of DN25 ball valve and be located to:
    - Allow sampling of air within the inlet plenum of the carbon vessel;
    - Allow sampling of air at 25% depth of the activated carbon bed;
    - Allow sampling of air at 50% depth of the activated carbon bed;
    - Allow sampling of air at 75% depth of the activated carbon bed; and
    - Allow sampling of air within the outlet plenum of the carbon vessel.
  - 2) All sampling ports along the AC vessel shall be fitted with devices which prevent aerosols or AC media from exiting the port when opened.
- (f) Utility
  - (i) Plant fire service water or comparable quality plant service water shall be the only acceptable source of water for the fire-fighting sprinkler system sprays to the individual AC beds
  - (ii) Dedicated safety showers with eye wash stations shall be provided near the AC vessels, with a maximum distance of 6 m between safety showers. Potable water shall be the only acceptable source of water for the safety showers and eye wash stations.
  - (iii) Dedicated hose reels shall also be provided in the vicinity of the pre-filters, and shall be accessible to all AC vessels, which may be sourced by potable water or plant service water, if available on site.
- (g) AC Regeneration System
  - (i) Should regenerable AC (GAC) be proposed by the Contractor, the following requirements must be satisfied:
    - 1) Full details of the GAC re-generation system shall be provided with the tender response; and
    - 2) The type of the regenerating process and reagents required, frequency and duration of the re-generation cycle and volume of waste produced during each cycle shall be provided with the tender response.
  - (ii) Drainage facilities for the excessive/waste re-generating liquid shall be fitted with water traps and water seals to ensure no simultaneous escape of odorous air.
- (h) Spent Media Disposal
  - (i) The method of removal, disposal, and waste classification of the spent media shall be stated in the tender submissions.
  - (ii) The time required to completely remove all the spent media from each treatment unit and replenish it with fresh media shall be stated in the tender documents.

#### **G4.72.10 Bio-Trickling Filter (BTF) Design Requirements**

(a) General

- (i) This section covers odour control equipment where the treatment process utilises the growth of a biofilm on an inert media irrigated with final effluent or potable water supplemented with nutrients, such as a Bio-Trickling Filter.
- (ii) The irrigating liquid, type, quality, pressure, instantaneous flow demand, and total daily irrigating liquid consumption shall be stated in the Contractor's tender submissions.
- (iii) The target operating pH of the BTF shall be stated in the Contractor's tender submissions.
- (iv) Each BTF shall comprise the following as a minimum:
  - 1) BTF vessel internals including media, media supports, media liquid irrigation system, and a mist eliminator integral to the BTF vessel;
  - 2) Irrigating liquid conditioning system, inclusive of nutrient storage tanks, transfer pumps, piping, valves, and nutrient flow monitoring devices (Contractor and system dependant);
  - 3) Irrigating liquid irrigation system, inclusive of irrigation water tanks, irrigation pumps, piping, valves, and irrigation flow monitoring devices (Contractor and system dependant);
  - 4) Irrigating liquid re-circulating system (Contractor and system dependant):
    - If a recirculating irrigation liquid system is used, the BTF system shall include a liquid recirculating sump with associated pumps, automated level control and associated level indicators;
    - If a recirculating liquid irrigation system is used, an automated process shall also be included to periodically or continuously purge the liquid recirculating sump to a drain to prevent the accumulation of reaction products in the BTF blowdown; and
    - A sump overflow line shall also be used in the event of a level control system failure to prevent the tower from filling with liquid and flooding the air inletline.
  - 5) BTF irrigation liquid blowdown system (Contractor and system dependant).
- (v) The complete BTF system as proposed by the Contractor shall fit within the designated areas as outlined in the Particular Specifications.
  - 1) The vessel dimensions (length x width and/or diameter) and media volume shall be calculated by the Contractor and properly designed to achieve the specified Performance Guarantees outlined in the Particular Specifications.

- 2) The following minimum clearance distances between adjacent BTF vessels shall be satisfied to allow adequate access for maintenance:
- Minimum of 1.5 m between a set of two (2) adjacent Bio-Trickling Filters (BTFs);
  - Minimum of 500 mm clearance between ducts and BTF vessels unless it is walkway, in which case a minimum of 750 mm clearance shall be provided between ducts and BTF vessels; and
  - All BTF vessels shall be installed adjacent to a dedicated service road or grassed area capable of passing standard sized vehicles and trucks along one side of the vessel.
- (vi) Unless otherwise specified, the installation shall only incorporate bypass ductwork to allow odorous air to be removed from its source with all BTF units offline when the BTF system is combined with an AC scrubber polishing system.
- (vii) Detail of the number, treatment stages and configuration of the BTFs and ancillary components/system provided shall be stated by the Contractor in the tender submissions and satisfy the minimum Empty Bed Residence Time (EBRT or contact time) as outlined in the Particular Specifications .
- (viii) Each treatment stage shall comprise a random or structured treatment bed of inert media and its associated support structure. The volume and arrangement of the media shall be such that the rates of mass transfer of odorants from the gas to the liquid phase within each treatment stage are optimised to achieve the required performance.
- (ix) The BTF system inclusive of its irrigation system, nutrient supplement system, recirculation system (if supplied) and blowdown system (if supplied) shall be designed and constructed such that the system is fully automated with minimal operator intervention.
- (x) The manufacturer shall advise the alarm and interlock settings for the safe and reliable operation of the BTF system. The following interlocks and alarms must be built into the Odour Treatment Plant control system and local operator interface, in addition to the standard alarms required for operation of the entire OCF:
- 1) For each BTF vessel:
- Low irrigation flow alarm;
  - High bed differential pressure alarm;
  - BTF internal sump high level alarm;
  - BTF liquor discharge low pH alarm; and
  - BTF liquor discharge high pH alarm.
- (xi) All pump wetted parts shall be corrosion resistant taking into consideration the pumped fluid, concentration of any chemical present and the temperature.

(b) BTF Media Requirements

- (i) The inert media shall be constructed of polyurethane foam with a minimum five-year proven track record of operation under similar operating conditions as that stipulated in this tender in the Specifications. Alternative media such as engineered synthetic media and lava rock may be considered if a suitable track record under similar operating pollutant loads can be demonstrated. Plastic media from polypropylene, polyethylene, or other non-porous plastic materials such as pall ring media are not acceptable.
- (ii) Random or structured media are both acceptable. If random packed media is used, the Contractor shall limit the depth of each media layer to 1.5 m to reduce the risk of media compaction over time.
- (iii) The proposed media shall have high porosity and shall not shrink or swell with varying moisture content.
- (iv) The proposed inert media shall not inhibit the growth of a biofilm nor have any leachable constituents, which could be detrimental to the operation of the treatment works or receiving waters.
- (v) The inert media shall be compatible with and/or resistant to the scrubbing liquid and have sufficient compressive strength to resist all operational loads (its own weight and that of the entrained liquid and biomass at the installed depth) without crushing or deformation.
- (vi) The inert media shall be constructed to prevent short circuiting of the air being treated and selected to minimise the pressure loss across the scrubber vessel.
- (vii) The inert media shall be supported on a support grid provided with sufficient free area to allow the free passage of air up through it with the simultaneous drainage of irrigation water and sloughed biomass down through it.
- (viii) The volume of inert media provided in the BTF shall be sufficient to achieve the required contact time with the media to achieve the reduction in H<sub>2</sub>S and odour as specified in the Particular Specification.
- (ix) The following media information shall be stated by the Contractor in the tender submissions as a minimum, and are outlined in the Particular Specifications:
  - 1) The type and composition of media;
  - 2) Specific surface area; and
  - 3) Pressure drop across the media at given air velocities.
- (x) The inert media shall be designed such that it can be readily removed and reinstalled for inspection, cleaning, or replacement. The minimum time between replacement of media shall be ten (10) years.
- (xi) The method of removal, disposal, and waste classification of the spent media shall be stated in the tender submissions.



- (xii) The time required to completely removed all the spent media from each treatment unit and replenish it with fresh media shall be stated in the Contractor's tender submissions.
- (c) BTF Irrigation System
  - (i) An irrigation system shall be supplied to provide uniform distribution of the scrubber irrigation liquid over the surface of the media.
  - (ii) Two sources of water for makeup of the BTF irrigation liquid is required, with one being from potable water as an emergency backup. The BTF irrigation liquid quality requirements shall be stated on the Contractor's tender submissions
  - (iii) Full details of the BTF irrigation liquid nutrient makeup system, BTF irrigation liquid system, BTF recirculating system (if required for long term permanent normal operation of the BTF system) and BTF irrigation liquid blowdown (if provided), shall be provided with the Contractor's tender submissions.
  - (iv) The BTF irrigation liquid blowdown system (if supplied) shall regulate the levels of dissolved oxidation products in the irrigating liquid and thereby maintain its target pH. If supplied, full detail of the blowdown system shall be provided with the Contractor's tender submissions. The BTF irrigation liquid blowdown system (if supplied) shall incorporate facilities to visually confirm there is a flow of spent scrubbing liquid to drain.
  - (v) The waste BTF irrigation liquid characteristics shall be stated on the Contractor's tender submissions.
  - (vi) The BTF irrigation supply system shall incorporate the following as minimum:
    - 1) Irrigation water supply tanks, complete with dedicated duty and standby irrigation supply pumps, associated controls, piping, fittings, valves, and supports, for each water supply tank as a minimum.
    - 2) The proposed irrigation system shall consist of spray nozzles located above the packing media. Nozzles shall be of a flanged non-clogging type, designed to minimise headloss and arranged to ensure that the entire face of the media bed receives a uniform distribution of irrigating liquid. Nozzles shall be constructed of PVC or polypropylene. The arrangement of each spray bar and nozzle shall be such that it is minimises pipework and is accessible and can be readily removed from the BTF tower without the need for personnel to physically enter the tower or remove any ductwork. Confirmation of adequate spray pattern to be provided in factory acceptance tests.
    - 3) The irrigation water supply tank shall provide a minimum of four (4) hours storage at the average BTF irrigation water flow demand. The irrigation water supply tank shall be installed outdoors as part of the OCF package, and shall be suitable for exposure to ambient conditions as outlined in the relevant specifications.

- 4) Penetrations into the irrigation water supply tank wall shall be manufactured of HDPE or FRP fittings which shall be mounted onto the tank wall to ensure no leaks occur.
- 5) The system shall be furnished as a package and include the following equipment items as a minimum:
  - Overflow nozzle, level element nozzle, fill nozzle, and drain nozzle complete with valves;
  - Ergonomically designed inspection hatch fitted with viewing panels. for routine inspections complete with latch locks; and
  - All necessary controls, nozzles, and accessories to form a standalone system, including backflow preventers (RPZD) on the potable water line take-off (if provided), associated non-return valves, automated inlet supply valves, and hose reels.
- (vii) Wire mesh strainer on the irrigation water line from the irrigation water storage tank - to prevent solids carry over and blocking of the irrigation system, such as spray nozzles. The strainer mesh size shall be selected based on the spray nozzles size and specifications proposed by the Contractor.
- (viii) A flow monitoring and transmitting device shall be provided for measurement of the instantaneous BTF irrigation flow.
- (ix) A BTF irrigation water cabinet panel comprising of a centralised facility where all the BTF irrigation motorized valves are co-located, to enable easy access for daily monitoring and control, for once-through systems.
  - 1) The water cabinet panel shall be mounted on a support frame suitable for field installation;
  - 2) The panel shall be designed for outdoor use and shall have IP 65 protection;
  - 3) The panel shall contain isolation valves, control valves, pressure regulating valves, a flow meter, and all plumbing required for irrigation control for the BTF media;
  - 4) The panel shall include flanged connections, a raised face complying with BS EN 1092 PN 16, ability for field connections to external water supply, and discharge water piping to the BTF; and
  - 5) The Contractor shall provide details of the irrigation water panel system, which include but are not limited to P&ID drawings and operating control philosophies for the S.O.'s approval.
- (x) A pressure relief device and irrigation liquid overflow from the BTF irrigation supply pumps – in the event of excess pressure/flow of irrigation liquid.
- (xi) An irrigation liquid and spent leachate sampling facility – to allow measurement of leachate pH for confirmation of BTF operation.
- (xii) The irrigating liquid requirement (i.e. type, quality, pressure, instantaneous flow, and total daily water demand) and the frequency and duration of the irrigation cycle shall be stated in the tender submissions.

- (xiii) Should recirculation of the irrigation liquid be required for operation of the BTFs, associated recirculation pumps, piping, fittings, valves, instrumentation, controls, and supports shall be provided. Makeup water (if required) shall be introduced to the system in a manner to prevent short circuiting to the drain.
- (d) Biotrickling Filter Supplemental Nutrient System
  - (i) A backup BTF nutrient supply facility consisting of a nutrient feed system shall be provided. The system shall be furnished as a package, complete with the following components as a minimum:
    - 1) Storage vessel complete with mixer, overflow nozzle, level element nozzle, fill nozzle, and two nozzles for installation of nutrient pump suction lines. The storage vessel shall be capable of being bolted down to comply with BS EN 13121-3 and BS EN 13923;
    - 2) Ergonomically designed hinged inspection hatches for routine inspections and nutrient batch makeup, complete with latch locks;
    - 3) Dedicated metering pumps, with one standby unit as a minimum, complete with associated controls, piping, fittings, valves, and supports;
    - 4) A calibration facility for the nutrient dosing pumps – to ensure that the correct quantity of nutrients is being delivered; and
    - 5) All necessary instrumentation, controls, and accessories to form a standalone system.
  - (ii) The supplemental nutrient feed system shall be installed outdoors as part of the OCF package, and shall be suitable for exposure to ambient conditions as outlined in the relevant specifications.
  - (iii) The adopted BTF nutrient shall be non-hazardous and non-toxic, and stated as such on its Materials Safety Data Sheet (MSDS).
- (e) Mist Eliminators
  - (i) Mist eliminators shall be installed inside the BTF vessel above the final media bed to prevent the carryover of liquid droplets with the outlet gas stream, and satisfy the following requirements as a minimum:
    - 1) Remove at least 90 percent of entrained droplets greater than 10 microns in diameter, and at least 99 % of all droplets greater than 40 microns in diameter;
    - 2) It shall be constructed from polypropylene or PVC;
    - 3) It shall be configured and constructed to allow ease of disassembly and removal; and
    - 4) It shall not have greater than 100 Pa pressure drop across the entire mist eliminator depth. Sample lines and pressure gauges shall be provided in accessible locations for monitoring of the pressure drop across each BTF mist eliminator.

- (f) Specific Design Requirements for BTF
  - (i) All materials that will come into contact with the irrigation liquid shall be compatible with and/or resistant to the irrigation liquid. The minimum pH value that the materials are designed to withstand, without excessive corrosion, shall be stated in the Contractor's tender submissions.
  - (ii) A differential pressure indicator across the inlet and outlet of each BTF vessel, with an additional measurement port at the inlet of the BTF mist eliminator section, shall be provided to monitor the pressure drop through the media and mist eliminators in the vessels. Dedicated sample ports shall be provided at these lines to allow collection of samples from the BTF. All sample ports shall be piped to ground level.
  - (iii) The BTF housing shall be designed to contain the bed of media above a recycle liquor sump and air inlet plenum. An outlet air plenum and mist eliminator shall also be provided above the bed of media. External mist eliminators in the ductwork separate from the BTF will not be permitted.
  - (iv) The blowdown line from each BTF vessel shall be fitted with "U" or "S" bend water traps to minimise the likelihood of untreated odorous air escaping from the BTF vessels. The blowdown line from each BTF vessel will be connected to a common drain line or blowdown sump (if supplied). The "U" or "S" bend shall satisfy the requirements outlined in G4.72.8(b)(iii).

#### **G4.72.11 Chemical Scrubber Design Requirements**

- (a) General
  - (i) This section covers odour control equipment where the treatment process utilises chemical scrubbing to remove contaminants from the foul air stream. The wetted scrubber packing enables contact between the gas and the scrubbing chemicals and allows for removal of the contaminants from the foul air stream.
  - (ii) The irrigating liquid, type, quality, pressure, instantaneous flow demand, and total daily irrigating liquid consumption (inclusive of chemical consumption) shall be stated in the Contractor's tender submissions.
  - (iii) The target operating pH and ORP of the Chemical Scrubber shall be stated in the Contractor's tender submissions.
  - (iv) No exposed metal items shall come in contact with any process air or liquid streams.
  - (v) Certified lifting lugs shall be designed, certified and provided on each Chemical Scrubber vessel segment and any equipment requiring crane assembly / disassembly for maintenance or repair. Each lifting point shall be labelled with its WLL.
  - (vi) The Chemical Scrubbers shall receive treated air through a single flanged inlet on the base of the vessel. The inlet plenum shall be designed to ensure even flow distribution through the scrubber media.
  - (vii) The Chemical Scrubber shall discharge treated air through a single flanged outlet on top of the vessel via the outlet duct.

- (viii) Each Chemical Scrubber shall comprise the following features as a minimum:
- 1) The Chemical Scrubber shall be a vertical counter-current tower fitted with all necessary internals to provide a complete functional system inclusive of recirculation sump and pump station.
  - 2) Chemical Scrubber tower internals are to include scrubber media, media supports, liquid irrigation system for the media, and a mist eliminator integral to the scrubber vessel.
  - 3) Mist eliminators are to be fitted internally at the top of the scrubber, before the outlet duct connection. A cleaning irrigation spray system shall also be provided for intermittent cleaning of the mist eliminator.
  - 4) The vessel diameter and media volume shall be calculated by the Contractor and properly designed to achieve the specified performance criteria. The design shall minimise the gas side differential pressure across the tower. The vessel diameter shall not exceed that nominated in the particular specification.
  - 5) The scrubber media irrigation system shall be in the form of a recirculation system with spray nozzles designed to ensure complete wetting of the media. Recirculation liquor sump shall be integral to the scrubber vessel (Contractor and system dependant).
  - 6) Irrigating liquid conditioning system, inclusive of chemical (sodium hypochlorite and sodium hydroxide) dosing skids complete with dosing pumps, piping, valves, and chemical flow monitoring devices (Contractor and system dependant).
  - 7) The Chemical Scrubbers liquid recirculation system shall include:
    - A liquid recirculation sump with associated pumps, piping, valves, spray nozzles, irrigation flow monitoring devices, sump automated level control and associated level indicators.
    - Provision of duty/standby recirculation pumps with low flow protection and automated cut-over between the pumps.
    - An automated process shall be included to fill/top up the recirculation sump with fresh water.
    - An automated process shall also be included to periodically or continuously purge the liquid recirculating sump to a drain to prevent the accumulation of reaction products.
    - Scrubbing chemicals shall be dosed to the recirculation liquor via inline static mixers downstream of the recirculation liquor flowmeter.
    - Mist eliminator cleaning irrigation system.

- The take-off point for the online continuous sampling and monitoring of the recirculation liquor pH and ORP shall be connected on the discharge side of the recirculation pump, prior to the flowmeter. The sampling line contents shall be returned to the sump.
  - A sump overflow line shall also be provided for the event of a level control system failure. This overflow will prevent the tower from filling with liquid and flooding the inlet foul air duct.
- 8) Chemical Scrubber irrigation liquid purge system (Contractor and system dependant)
- 9) Access ports shall be provided for each spray assembly (mist eliminator and scrubber media).
- (ix) The complete Chemical Scrubber system inclusive of all components as proposed by the Contractor shall fit within the designated areas as outlined in the Particular Specifications.
- 1) The vessel dimensions and media volume shall be calculated by the Contractor and properly designed to achieve the required Performance Guarantees outlined in the Particular Specifications.
- 2) The following minimum clearance distances between adjacent Chemical Scrubber vessels shall be satisfied to allow adequate access for maintenance:
- Minimum of 1.5 m between the two (2) adjacent Chemical Scrubbers.
  - Minimum of 500 mm clearance between ducts and Chemical Scrubber vessels unless it is a walkway, in which case a minimum of 750 mm clearance shall be provided between ducts and vessels.
  - Chemical Scrubber vessels shall be installed adjacent to a dedicated service road or grassed area capable of passing standard sized vehicles and trucks along one side of the vessel.
- (x) Detail of the number and configuration of the Chemical Scrubbers and ancillary components/system provided shall be stated by the Contractor in the tender submissions and satisfy the minimum requirements outlined in the Particular Specifications .
- (xi) Each Chemical Scrubber shall comprise treatment beds of inert media and its associated support structure. The volume and arrangement of the media shall be such that the rates of mass transfer of odorants from the gas to the liquid phase are optimised to achieve the required performance.
- (xii) The Chemical Scrubber system inclusive of its irrigation system (nozzles and spray bars), chemical dosing systems, recirculation systems and purge systems shall be designed and constructed such that the system is fully automated with minimal operator intervention.

- (xiii) The Manufacturer shall advise the alarm and interlock settings for the safe and reliable operation of the Chemical Scrubber system. The following interlocks and alarms must be built into the Odour Treatment Plant control system and local operator interface, in addition to the standard alarms required for operation of the entire OCF:

1) For each Chemical Scrubber vessel:

- Recirculation liquor low flow alarm;
- High media differential pressure alarm;
- High mist eliminator differential pressure alarm;
- Scrubber sump low level alarm;
- Scrubber sump high level alarm;
- Scrubber recirculation liquor low pH alarm;
- Scrubber recirculation liquor high pH alarm;
- Scrubber recirculation liquor low ORP alarm;
- Scrubber recirculation liquor high ORP alarm; and
- Scrubber pH/ORP probe deviation alarm.

- (xiv) All pump wetted parts shall be corrosion resistant taking into consideration the pumped fluid, concentrations of chemicals present and the temperature.

(b) Scrubber Packing Requirements

- (i) The scrubber packing media shall be constructed of synthetic material that is compatible with the reagents (sodium hypochlorite and sodium hydroxide), foul air products of reaction and citric acid (used for cleaning purposes) with a minimum five-year proven track record of operation under similar operating conditions as that stipulated in this tender in the Specifications.
- (ii) The proposed packing media shall have high porosity and shall not shrink or swell with varying moisture content.
- (iii) The packing media is to be randomly placed.
- (iv) The packing media shall be selected to prevent short circuiting of the air being treated and to minimise the pressure loss across the surface of the scrubber vessel.
- (v) The packing media shall be biologically and chemically resistant to the liquid and gas stream operating conditions and contaminants.
- (vi) The inert packing media shall have sufficient compressive strength to resist all operational loads (its own weight and that of the entrained liquid at the installed depth) without crushing or deformation.
- (vii) The inert packing media shall be supported on a support grid provided with sufficient free area to allow the free passage of air up through it with the simultaneous drainage of irrigation liquor down through it.

- (viii) The volume of inert packing media provided in the Scrubber shall be sufficient to achieve the required contact time with the media to achieve the reduction in H<sub>2</sub>S and odour as specified in the Particular Specification.
  - (ix) The following media information shall be stated by the Contractors in the tender submissions as a minimum, and are outlined in the Particular Specifications:
    - 1) The type and composition of media;
    - 2) Specific surface area; and
    - 3) Pressure drop across the media at given air velocities.
  - (x) The inert packing media shall be designed such that it can be readily removed and reinstalled for inspection, cleaning, or replacement. The minimum time between replacement of media shall be five (5) years.
  - (xi) Any specific packing media cleaning requirements (e.g. acid cleans) including the frequency of cleaning, method of cleaning, and time required for cleaning, shall be stated in the Contractor's tender submissions.
  - (xii) The method of removal, disposal, and waste classification of the spent media shall be stated in the tender submissions.
  - (xiii) The time required to completely remove all the spent media from each treatment unit and replenish it with fresh media shall be stated in the Contractor's tender submissions.
- (c) Scrubber Irrigation System
- (i) An irrigation system shall be supplied to provide uniform distribution of the scrubber irrigation liquid over the surface of the media (during normal operation), and the mist eliminator (for maintenance purposes).
  - (ii) Full details of the Scrubber irrigation systems (media and mist eliminator), chemical dosing systems, recirculating systems, and the scrubber irrigation liquid purging systems shall be provided with the Contractor's tender submissions.
  - (iii) The proposed Chemical Scrubber irrigation systems shall consist of:
    - 1) Spray nozzles located above the packing media, for irrigation of the media; and
    - 2) Spray nozzles located above the mist eliminator, for intermittent cleaning of the mist eliminator.
  - (iv) Spray nozzles for both the media irrigation and mist eliminator cleaning shall be of a flanged non-clogging type, designed to minimise headloss and arranged to ensure that the entire face of the irrigated surface (packing media or mist eliminator pads) receives a uniform distribution of irrigating liquid.



- (v) Spray nozzles shall be constructed of PVC or polypropylene. The arrangement of each spray bar and nozzle shall be such that it minimises pipework, is easily accessible, and can be readily removed from the scrubber vessel without the need for personnel to physically enter the vessel or remove any ductwork. Confirmation of adequate spray pattern shall be provided in factory acceptance tests.
  - (vi) The scrubber irrigation liquor purging system shall regulate the build-up of dissolved salts in the scrubber liquor. Full detail of the purging system shall be provided with the Contractor's tender submissions. The scrubber irrigation liquor purging system shall incorporate facilities to visually confirm there is a flow of spent scrubbing liquid to drain. **Note: conductivity monitoring may also be considered for control of purging.**
  - (vii) The purged irrigation liquor characteristics shall be stated on the Contractor's tender submissions.
- (d) Scrubber Irrigation Water Supply System
- (i) The scrubber irrigation water supply system provides fresh water to the chemical scrubbers and shall incorporate the following as minimum:
    - 1) Irrigation water supply tank, complete with dedicated duty and standby irrigation water supply pumps, associated controls, piping, fittings, valves, and supports, as a minimum.
    - 2) The irrigation water supply tank shall provide a minimum storage capacity of the total chemical scrubber irrigation liquor sump volumes. The irrigation water supply tank shall be installed outdoors as part of the OCS package, and shall be suitable for exposure to ambient conditions as outlined in the relevant specifications.
    - 3) Penetrations into the irrigation water supply tank wall shall be manufactured of HDPE or FRP fittings which shall be mounted onto the tank wall to ensure no leaks occur.
    - 4) The irrigation water supply tank system shall be furnished as a package and include the following equipment items as a minimum:
      - Overflow nozzle, level element nozzle, fill nozzle, and drain nozzle complete with valves;
      - Ergonomically designed inspection hatch fitted with viewing panels. for routine inspections complete with latch locks; and
      - All necessary controls, nozzles, and accessories to form a standalone system, including backflow preventers (RPZD) on the potable water line take-off (if provided), associated non-return valves, automated inlet supply valves, and hose reels.
  - (ii) Wire mesh strainer on the irrigation water line - to prevent solids carry over and blocking of the irrigation system, such as spray nozzles. The strainer mesh size shall be selected based on the irrigation spray nozzles size and specifications proposed by the Contractor.
  - (iii) A pressure monitoring device for the scrubber irrigation water supply line.

- (iv) A pulsation dampener on the irrigation water supply pump combined outlet line to minimise pump stops and starts.
  - (v) The irrigating water supply requirement (i.e. type, quality, pressure, instantaneous flow, and total daily water demand) shall be stated in the tender submissions.
- (e) Chemical Scrubber Chemical Dosing Systems
  - (i) Chemical storage tanks as defined in G4.47 Chemical Plant, and shall accommodate a minimum thirty (30) days storage capacity of sodium hydroxide and sodium hypochlorite at average H<sub>2</sub>S loads at peak airflows to the Chemical Scrubbers.
  - (ii) A minimum of two (2) storage tanks shall be provided for each chemical, to provide the minimum required storage capacity.
  - (iii) A sodium hypochlorite dosing skid for each Scrubber shall be provided to maintain the scrubbing liquor ORP levels. The system shall be furnished as a package, complete with the following components as a minimum:
    - 1) Dedicated metering pumps, with one standby unit as a minimum, complete with associated controls, piping, fittings, valves, and supports.
    - 2) A calibration facility for the dosing pumps – to ensure that the correct quantity of sodium hypochlorite is being delivered.
    - 3) All necessary instrumentation (inclusive of flowmeters), controls, and accessories to form a standalone system.
  - (iv) A sodium hydroxide dosing skid for each Scrubber shall be provided to maintain the scrubbing liquor pH levels. The system shall be furnished as a package, complete with the following components as a minimum:
    - 1) Dedicated metering pumps, with one standby unit as a minimum, complete with associated controls, piping, fittings, valves, and supports.
    - 2) A calibration facility for the dosing pumps – to ensure that the correct quantity of sodium hydroxide is being delivered.
    - 3) All necessary instrumentation (inclusive of flowmeters), controls, and accessories to form a standalone system.
  - (v) The chemical dosing systems shall be installed outdoors as part of the OCS package, and shall be suitable for exposure to ambient conditions as outlined in the relevant specifications.
- (f) Scrubber Mist Eliminators
  - (i) Mist eliminators shall be installed above the scrubber packing media to prevent the carryover of liquid droplets with the outlet gas stream, and satisfy the following requirements as a minimum:
    - 1) Remove at least 95 % of entrained droplets greater than 10 microns in diameter, and at least 99 % of all droplets greater than 40 microns in diameter.

- 2) It shall be constructed from polypropylene or PVC.
  - 3) It shall be configured and constructed to allow ease of disassembly and removal.
  - 4) It shall not have greater than 100 Pa pressure drop across the entire mist eliminator depth. Sample lines and pressure gauges shall be provided in accessible locations for monitoring of the pressure drop across each scrubber mist eliminator.
- (g) Specific Design Requirements for the Chemical Scrubbers
- (i) All materials that will come into contact with the irrigation liquid shall be compatible with and/or resistant to the irrigation liquor which consists of sodium hypochlorite, sodium hydroxide and citric acid. The minimum and maximum pH values that the materials are designed to withstand, without excessive degradation, shall be stated in the Contractor's tender submissions.
  - (ii) A differential pressure indicator across the inlet and outlet of each Scrubber vessel, with an additional measurement port at the inlet of the Scrubber mist eliminator section, shall be provided to monitor the pressure drop through the media and mist eliminators in the vessels. Dedicated sample ports shall be provided at these lines to allow collection of samples from the Scrubbers. All sample ports shall be piped to ground level.
  - (iii) The Scrubber housing shall be designed to contain the bed of media above the recirculation liquor sump and air inlet plenum. An outlet air plenum and mist eliminator shall also be provided above the bed of media. External mist eliminators in the ductwork separate from the Scrubber will not be permitted.
  - (iv) The purge line from each Scrubber vessel shall be fitted with "U" or "S" bend water traps to minimise the likelihood of untreated odorous air escaping from the vessels. The purge line from each scrubber vessel will be connected to a common drain line. The "U" or "S" bend material requirements shall satisfy the requirements outlined in G4.72.8(c)(ii).

#### **G4.72.12 Biofilter Design Requirements**

- (a) General
- (i) This section covers odour control equipment where the treatment process utilises the growth of a biological microorganisms suspended on engineered biofilter media, or traditional solid media systems, such as compost, mulch, wood chips, bark.
  - (ii) Each Biofilter shall comprise the following as a minimum:
    - 1) Biofilter vessels (nominally reinforced concrete) with corrosion resistant plastic liner or coating, compatible with a pH of between 1.5 and 8.5;
    - 2) Biofilter pre-humidification system for primary moisture control of the Biofilter media;

- 3) Biofilter inlet plenum and baseplate air distribution plenum system to provide even airflow distribution across the Biofilter media;
  - 4) Biofilter media, which include but are not limited to: engineered media, soil bed media, woodchips, mulch, scoria, and similar. The type of media required is outlined in the Particular Specifications;
  - 5) Biofilter irrigating system for secondary moisture control of the Biofilter media, inclusive of irrigation booster pumps (if required), air break tanks and pumping systems (for irrigation using potable water), irrigation piping and irrigation sprays, motorised valves, and flow monitoring devices (dependant on the irrigation liquid used);
  - 6) Biofilter drainage piping and disposal system at each Biofilter cell, fitted with water traps to prevent accidental egress of untreated foul air;
  - 7) Portable sampling flux hoods, at least one for each Biofilter Cell.
- (iii) All materials that will come into contact with the Biofilter condensate shall be compatible with and/or resistant to the condensate. The minimum pH value that the materials are designed to withstand, without excessive corrosion, shall be stated in the Contractor's tender submissions.
- (iv) The complete Biofilter system as proposed by the Contractor shall fit within the designated areas as outlined in the Particular Specifications.
- (v) The nominal number of cells, area per cell, cell dimensions (length x width), total area, including media depth and volume shall be calculated by the Contractor and stated in their tender submissions. The Biofilters shall be properly designed to achieve the specified minimum Empty Bed Residence Time (EBRT or contact time) and Performance Guarantees outlined in the Particular Specifications. This arrangement is intended to represent the minimum standard of quality and treatment capacity.
- (vi) The Biofilters shall be provided with a minimum clearance of four (4) m along at least three (3) sides of the Biofilters to facilitate access for media removal and replacement using bobcats, excavators, vacuum trucks, cranes, and similar heavy equipment.
- (vii) The Biofilter installation shall not incorporate any bypass ductwork to allow untreated odorous air to be directly discharged to the environment.
- (viii) The Biofilter system inclusive of its humidification and irrigation system shall be designed and constructed such that the system is fully automated with minimal operator intervention.
- (ix) The Manufacturer shall advise the alarm and interlock settings for the safe and reliable operation of the Biofilter system. The following interlocks and alarms must be built into the control system and local operator interface, as a minimum, in addition to the standard alarms required for operation of the entire OCS:
- 1) For each Biofilter Cell:
    - Low pre-humidification flow alarm;
    - Low irrigation flow alarm;

- Biofilter distribution plenum high level alarm.
- (x) Each Biofilter cell shall be provided with a portable flux hood, which shall be manufactured from clear Plexiglass or Fibreglass Reinforced Plastic. The hood shall be at least 0.2m<sup>2</sup> area and 300mm deep. The bottom of the flux hood shall remain open, and the top of the flux hood shall be provided with a DN25 connection.
- (xi) Wire mesh strainers shall be provided on the Biofilter humidification and irrigation water line take-off from plant water to prevent solids carry over and blockages of the humidification and irrigation system components, such as spray nozzles. The strainer mesh size shall be selected based on the spray nozzles size and specifications proposed by the Contractor and outlined in the Particular Specifications.
- (xii) If the Biofilter irrigation and humidification water supply system is to be supplied from potable water, an air gap system shall be provided to provide the separation requirement to avoid cross contamination. In this instance, the Contractor shall provide and install all components considered necessary for the air gap system to function properly. The required components include the following but is not limited to: an air gap tank, air gap potable water float valve, fiberglass enclosure, and the humidification and irrigation pump.
- (xiii) Flow meters with transmitting capabilities shall be provided, one to measure the total instantaneous Biofilter Irrigation water flow and one to measure the total instantaneous Biofilter Humidification water flow. Local flow indicators shall also be provided to measure the humidification water flow to each Humidification zone, and the instantaneous irrigation water flow to each Biofilter zone as indicated on the Drawings.
- (xiv) The leachate and condensate collection drain line from each Biofilter cell shall be fitted with “U” or “S” bend water traps filled with water to minimise the likelihood of untreated odorous air escaping from the Biofilter cells. The drain line from each Biofilter vessel will be connected to a common drain line or blowdown sump (if supplied). The “U” or “S” bend water traps shall satisfy the requirements outlined in G4.72.8(b)(iii).
- (b) Biofilter Media Requirements
  - (i) Finished surface of the media shall be made level and smooth. No equipment shall be allowed to be placed directly on top of the media at any time, and personnel shall not directly walk or stand on the media. Foot traffic shall be limited to access platforms across the Biofilter surface, or boards placed on top of the engineered media.
  - (ii) Biofilter media shall be stored to prevent damage to components, including the engineered air distribution plenum system and main header assembly.
  - (iii) The Biofilter cells shall be designed such that Biofilter media can be readily removed and reinstalled for inspection, cleaning, or replacement.
    - 1) The method of removal, disposal, and waste classification of the spent media shall be stated in the Contractor tender submissions.

- 2) The time required to completely removed all the spent media from each treatment unit and replenish it with fresh media shall be stated in the Contractor's tender submissions.
- (iv) The Contractor shall warrant the Biofilter media for the minimum period from the date of Substantial Completion as outlined in the Particular Specifications. This warranty shall be issued to The Board. In addition:
  - 1) The media shall not shrink or compact by more than five percent (5%) within six (6) months of installation, or more than ten percent (10%) of the original bed depth over its warranted life, as indicated in differential pressure drops exceeding maximum limits stated herein at the design airflow rates.
  - 2) The media shall not degrade or decompose during the warranty period and shall not shrink or swell with varying moisture conditions.
  - 3) The Biofilter shall meet the performance criteria specified in the Particular Specifications throughout the warranty period.
  - 4) If the media degrades or is shown to be defective in material or workmanship, or the performance criteria is exceeded, upon examination by the media manufacturer, the Contractor shall either:
    - Repair or replace any part of the defective media at no cost to The Board; or
    - Replace the entire media bed at no cost to The Board.
- (v) Adequate media depth shall be provided to supply the minimum empty bed residence time (EBRT) as outlined in the Particular Specifications. The total media bed depth shall not exceed the following parameters:
  - 1) Maximum one (1) meter height for soil bed media Biofilters;
  - 2) Maximum two (2) meters height for engineered media Biofilters.
- (vi) For engineered Biofilter media, the following requirements shall also apply:
  - 1) All media surfaces shall be one hundred percent (100%) coated. Plain uncoated media shall not be acceptable.
  - 2) Engineered biofilter media shall be formulated with the proper nutrients and buffering agents for its intended operation.
  - 3) Engineered biofilter media proposed for installation shall have a proven track record with at least THREE (3) comparable installations under comparable scale and service conditions (airflows, loads, and process guarantees). Each comparable installation shall have been operating successfully and satisfied its associated process guarantees for a minimum of five (5) years. References of installation for the proposed engineered biofilter media shall be provided as part of the Contractor's tender submissions.

- (c) Biofilter Baseplate Air Distribution Plenum
- (i) The inert media shall be supported on an engineered support grid consisting of a modular, self-contained corrosion resistant perforated floor system installed over a reinforced concrete slab as indicated on the Drawings issued in the Particular Specifications. Only specialised engineered perforated floor systems are acceptable.
  - (ii) The engineered perforated floor system shall conform to the following requirements:
    - 1) Airflow through the floor emitters shall be sized to provide a maximum face velocity of 0.05m/s, to distribute the required airflow to each Biofilter cell as defined.
    - 2) The baseplate floor system shall be capable of being accessed by support material handling equipment with vertical wheel loads of up to 127,000 kilograms per square meter (kg/m<sup>2</sup>).
    - 3) Trench cover panels capable of spanning not less than 750mm shall be provided as necessary. Panels shall be of perforated or non-perforated HDPE and designed with joints to fit with HDPE floor panels.
    - 4) The floor emitters and trench cover panels shall be either compression moulded polyethylene or injection melded from a polyethylene / polypropylene composite.
    - 5) The engineered floor system is essential to ensuring proper airflow distribution and adequate treatment through the Biofilter cells. For this reason, the Contractor shall ensure that the selection, design, and manufacture of the engineered Biofilter distribution floor systems are certified as acceptable and compatible by the specialist Biofilter media Manufacturer.
    - 6) The floor system shall allow for biofilter condensate drainage. Legs shall be provided such that the Biofilter condensate can flow freely along the floor surface and around the support legs. The Biofilter condensate drain off point shall be fitted with a “U” or “S” bend provided with a water seal to prevent unintended short circuiting of foul air for treatment.
    - 7) Solid end plates shall be provided for sealing the perimeter edge of the floor systems. End plates shall be manufactured of HDPE or HDPE/PP or type 316SS and shall be bolted to the concrete wall to provided proper anchorage.
    - 8) All hardware shall be Type 316SS.
  - (iii) The main header trench which feed the baseplate air distribution plenum shall be fabricated from Stainless Steel 316 sheet metal of at least 3mm thickness to support the required media depth without deflection or deformation. All seams shall be welded. Sides and bottom of the main header shall be bolted to the concrete walls and floor, respectively, with Type 316 Stainless Steel anchor bolts.

- (iv) The engineered baseplate air distribution plenum system shall be installed plumb and straight, parallel to existing walls. The following should be observed during installation:
  - 1) Where gaps exist between the engineered floor system and concrete walls of the Biofilter cells, secured closure pieces, as approved by the S.O., shall be provided and installed by the Contractor;
  - 2) The entire plenum floor system shall be installed as a modular system for ease of removal and replacement. No welding or other permanent means of connections shall be utilised; and
  - 3) Legs shall be trimmed as necessary to install the engineered floor system flush to the bottom of the concrete Biofilter floor.
- (d) Humidifiers
  - (i) The Biofilter system shall include a fully automated humidification system upstream of the Biofilters for primary moisture control of the Biofilter media using plant water (G2D).
  - (ii) The Biofilter humidification system shall distribute water evenly across the cross-section of the humidification chamber to humidify the Biofilter inlet air to greater than ninety percent relative humidity (>90% RH). The humidification system assembly shall satisfy the following requirements:
    - 1) The spray assembly shall include but is not limited to: all necessary spray nozzles, humidifier water piping, pressure indicators, flow switches, in-line strainers, valves, and flow indicators;
    - 2) The humidification spray nozzle shall be mounted in the humidification chamber as shown on the Drawings;
    - 3) All other components of the Humidification system (apart from the spray nozzles) such as the pressure indicators, in-line strainers, control valves, diaphragm control valves, and flow indicators shall be enclosed within a water control panel provided for the Biofilter Humidification water system;
    - 4) Spray nozzles within the humidification chamber shall be easily removable for inspection and maintenance without shutdown of the Biofilter cell or odour treatment system, or disassembly of the entire humidification piping system. A bolted access hatch fitted with gaskets shall be provided on where the spray nozzles are installed, for ease of inspection and maintenance of the spray nozzles. Oversized nozzles and unions shall be provided as necessary; and
    - 5) Spray nozzles shall provide fine spray atomisation in a hollow cone pattern using hydraulic pressure alone. Each nozzle shall be constructed of Stainless Steel 316 and be fitted with an integral strainer.



- (iii) A centralised Biofilter humidification water panel comprising of a centralised facility where all the Biofilter humidification control instruments and valves are co-located shall be provided, to enable easy access for daily monitoring and control. The Biofilter humidification water cabinet panel shall satisfy the following requirements as a minimum:
  - 1) The water cabinet panel shall be mounted on a support frame suitable for field installation and be resistant to UV degradation;
  - 2) The panel shall be fabricated of Type 316 Stainless steel, designed for outdoor use and shall have IP 65 protection;
  - 3) The panel shall contain all necessary isolation valves, flow control valves, pressure regulating valves, instrumentation, and all plumbing required for humidification control for the Biofilter inlet air;
  - 4) The panel shall include flanged connections, a raised face complying with BS EN 1092 PN 16, ability for field connections to external water supply, and discharge water piping for the Biofilter humidification lines. Drain lines with a manual isolation valve shall also be provided for each of the humidification lines to facilitate maintenance; and
  - 5) The Contractor shall provide details of the Biofilter humidification water panel system, which include but are not limited to P&ID drawings, functional descriptions, and control philosophies for the S.O.'s approval.
- (e) Biofilter Irrigation System
  - (i) The irrigating liquid requirement (i.e. type, quality, pressure, instantaneous flow, and total daily water demand) and the frequency and duration of the Biofilter irrigation cycles shall be stated in the Contractor tender submissions.
  - (ii) A fully automated surface irrigation system shall be provided for secondary moisture control of the Biofilter media using plant water, and shall be segregated into independent zones if necessary.
    - 1) Each Biofilter irrigation zone shall be operated sequentially, rather than simultaneously, to reduce instantaneous water and pumping demand.
    - 2) Electric motorised control valves shall be provided for each irrigation zone, to enable the required automation.
    - 3) Each Biofilter irrigation zone shall include the following components, but are not limited to: sprinkler heads, piping, flow indicators, and motorised control valves for automated irrigation.
    - 4) The sprinkler system shall consist of sprinkler heads which shall be at least DN100 full or part circle pop-up spray head.
    - 5) Piping shall be PVC pipe; 14 bars pressure rated upstream from controls, 11 bars downstream; solvent welded sockets.

- (iii) The indicative layout of the Biofilter irrigation system, and the number of Biofilter irrigation zones are shown on the Drawings provided. The Contractor is responsible for designing the irrigation system to ensure the entire Biofilter media bed will be adequately wetted to sustain the biomass growth. All irrigation piping shall be sloped back to a system drain valve as shown on the Drawings.
- (iv) A centralised Biofilter Irrigation water cabinet panel comprising of a centralised facility where all the Biofilter irrigation control instruments and valves are co-located shall be provided, to enable easy access for daily monitoring and control. The Biofilter irrigation water cabinet panel shall satisfy the following requirements as a minimum:
  - 1) The water cabinet panel shall be mounted on a support frame suitable for field installation and be resistant to UV degradation;
  - 2) The panel shall be fabricated of Type 316 Stainless Steel, designed for outdoor use and shall have IP 65 protection;
  - 3) The panel shall contain all necessary isolation valves, flow control valves, pressure regulating valves, instrumentation, and all plumbing required for irrigation control for the Biofilter media;
  - 4) The panel shall include flanged connections, a raised face complying with BS EN 1092 PN 16, ability for field connections to external water supply, and discharge water piping to the Biofilters. Drain lines with a manual isolation valve shall also be provided for each of the irrigation lines to facilitate maintenance; and
  - 5) The Contractor shall provide details of the Biofilter irrigation water panel system, which include but are not limited to P&ID drawings, functional descriptions, and control philosophies for the S.O.'s approval.
- (f) Specific Design Requirements for Biofilters
  - (i) Adequate panels of FRP grating at adequate lengths shall be provided at the vicinity of the Biofilter surface for personnel access to the top of the Biofilter media, such that personnel will not directly walk or stand on the Biofilter media.
  - (ii) Access stoplogs shall be provided, one for each Biofilter cell.
    - 1) The access stoplogs shall be at least four (4) meters long and shall span the full Biofilter wall height.
    - 2) The access stoplogs shall be fabricated of fibreglass reinforced plastic (FRP) manufactured with premium corrosion resistant vinyl ester resin.
    - 3) Access stoplogs shall be provided with lifting lugs to allow a crane to easily attach to and lift the access stoplogs when required for media replacement and maintenance.
- (g) Biofilter media installation shall satisfy the following requirements:
  - 1) Before depositing media, debris shall be removed from the space to be occupied by the media;

- 2) Biofilter media shall be placed as loosely as possible in the Biofilter and in a consistent manner to preserve porosity;
- 3) The finished surface of the Biofilter media shall be made level and smooth. This shall be achieved without any compaction of the media. No equipment is allowed on top of the placed media at any time and foot traffic shall be limited to boards placed on media; and
- 4) Media shall be placed to prevent damage to components, including the baseplate air distribution plenum system and main duct header assembly.

#### **G4.72.13 Mechanical Equipment Design Requirements**

(a) Ductwork

- (i) The Contractor shall be required to give one week's notice to the nominated S.O. representative of their intention to connect to the required ducting connections. The Contractor shall be aware that the odorous air ducting connections may be "live" during connections.
- (ii) All ducting which form common air headers going into or out of the OCS shall comply with the following:
  - 1) All ductwork within the OCS compound shall be supported with stainless-steel 316 L brackets as necessary;
  - 2) Unless otherwise specified in the Particular Specifications, the ducting design shall comply with the maximum ductwork air velocity requirement outlined below:

| Duct Sizes       | Max Duct Velocity (m/s) |
|------------------|-------------------------|
| Up to DN150      | 8.0                     |
| DN150 to DN500   | 10.0                    |
| DN500 to DN1000  | 12.0                    |
| DN1000 to DN2000 | 15.0                    |

- 3) The maximum pressure drop for the critical legs shall be less than 2 Pa/m. If the pressure losses in the odour ducting is higher than the maximum specified pressure drop, or if negative pressure is expected, special design considerations for the ductwork will be required and shall be included in the submitted proposal to S.O. for approval with P.E. endorsement;

- 4) All joints in ductwork / manifolds shall be flanged, bell and spigot, or butt joints and shall provide an effectively sealed joint that provides a smooth air-flow internally, whilst preventing egress of the conveyed medium, or ingress from outside. Joints shall be able to be readily installed with minimum disruption to adjacent ductwork or fittings and shall be able to be similarly dismantled where this may be required. Gaskets shall be full face and the material shall be compatible with humid and corrosive conditions of operation and shall be submitted to S.O. for approval;
- 5) Flexible connections shall be provided at the interface of all mechanical equipment items and along the ducting route to accommodate differential movement and prevent vibration transmission;
- 6) All ductwork shall be provided with flanged connections at its termination points;
- 7) Ductwork shall be securely fixed to rigid supports which are designed to allow ductwork expansion and shrinkage due to temperature without damaging the gel coat. All clamps and brackets etc. shall be made from material required in this Specification unless otherwise specified;
- 8) Ductwork shall be designed with adequate fall to prevent the accumulation of any condensate. Adequate number of drain points shall be provided at strategic locations for removal of condensate, and shall satisfy the requirements outlined in Section G4.72.8(b)(iii);
- 9) All necessary access and sampling ports for the connection of pressure gauges, air velocity sensors etc. and for sampling shall be provided and piped to ground level if it is at height. If feasible, access ports for flow measurement and balancing using a handheld anemometer shall be positioned in straight length of ductwork, at least 5 ductwork diameters away from dampers;
- 10) All ductwork shall be UV resistant and shall be of a colour approved by The Board prior to installation; and
- 11) Ducting between the extraction points and the fan inlet shall be constructed to withstand the full suction pressure of the fan during blocked conditions (both duct and pre-filter).

(b) Odorous Air Fans

- (i) Where fitted, centrifugal, axial or mixed flow fans shall comply with the following:
  - 1) Fans shall be secured using anti-vibration mountings;
  - 2) Depending on the location of the fan installed, special acoustic enclosures may be required. Integral temperature sensors shall be provided at the fans to prevent excessive heat build-up during operation. Dedicated ventilation fans may be required for cooling of fans installed within acoustic enclosures and shall be provided as part of the Contractor package;

- 3) Terminal units enclosed fans shall incorporate attenuation linings to reduce transmitted noise;
  - 4) A low airflow sensor shall be incorporated at the fan discharge to provide an alarm in case of fan failure;
  - 5) To ensure that no hazardous gases are present in the housing, the control system shall ensure that on start-up, the fan runs for a sufficient period to purge the ductwork prior. This period shall not be less than 30 seconds;
  - 6) Positioning of the fans will be dependent on the type of odour treatment technology adopted. The following requirements should be observed unless otherwise specified in the Particular Specifications:
    - For instances where only BTFs are adopted, the fan(s) are typically positioned upstream of the odour treatment vessels and push air through the equipment;
    - For instances where an AC system, Biofilters, or Chemical Scrubber systems are implemented, fan(s) can be positioned both upstream and downstream of the odour treatment scrubbers and push/draw air through the equipment; and
    - For instances where a combined system such as that consisting of a BTF with AC system or BTF with Chemical Scrubber system is implemented, fans is preferably positioned between the two treatment stages;
  - 7) After fabrication, the impellers shall be cleaned and Non-Destructive Tests (NDT) carried out to check for damage and cracks. A report shall be provided on the NDT carried out for each impeller. If any rework is required as a result of the NDT findings, repairs shall be carried out and NDT redone to show no faults remain;
  - 8) The impeller and shaft shall be assembled and balanced as per ISO 14694:2003 and ISO/DIS 21940-14;
  - 9) The impeller and shaft assembly shall not exceed radial and axial run out to achieve the balance class as per ISO 14694:2003 and ISO/DIS 21940-14; and
  - 10) The shaft shall not deflect beyond the point where the balance class becomes offset during operation.
- (ii) The fan housing and pedestal shall satisfy the following requirements:
- 1) For the Odorous Air Fans, integral access platform shall be attached to the fan pedestal to allow access to the fan bearing vibration monitors; and
  - 2) The fan housing enclosure shall be provided with a drain line with a “U” or “S” bend water trap and isolation ball valve, for manual removal of condensate. The drain line shall satisfy the requirements in Section G4.72.8(b)(iii).

- (iii) The fan motor shall satisfy the following requirements:
  - 1) The selected fan motor shall be capable of continuous reliable operation when the fan is operating at a duty point of 10 % greater than that specified in the Particular Specifications; and
  - 2) The fan motor shall be wired to terminal boxes located external to the fan casing for ease of maintenance.
- (iv) All gaskets and sealants, regardless of where they are used, shall satisfy the following requirements:
  - 1) Resistant to UV and other conditions that might reasonably be assumed to exist in the odour control facility, (e.g. high humidity conditions, bird droppings, cleaning chemicals, disinfectant, etc.); and
  - 2) If self-adhesive strips are applied, they should maintain their adhesive properties for the design life (25 years) without more than a 20 % loss of adhesion compared to a new strip.
- (v) The fans shall satisfy the following occupational health and safety requirements:
  - 1) Equipment shall be guarded in accordance with local regulations as outlined in the General Specifications on Workmanship Requirements;
  - 2) Guards shall be strong enough to withstand personnel and other loads during maintenance and inspection activities; and
  - 3) Appropriate workplace health and safety warning signs complying with local regulations as outlined in the General Specification G1.3 on Workmanship Requirements. Machine Guards shall be fastened to all equipment to protect against rotating parts.
- (vi) The fans shall satisfy the following requirements for noise:
  - 1) Tenders are to supply with their tender noise emission data (overall at 1 m free field and Octave band) to allow the S.O. to determine if dB (A) noise attenuation of the fans is required. Noise testing shall take place off-site and will be confirmed during commissioning on-site.
  - 2) Noise pressure levels shall not exceed the limits outlined elsewhere in the Contract.
  - 3) Where acoustic enclosures are to be provided by the Tenderer, they shall be designed, constructed and installed to ensure:
    - Normal fan operation is possible without opening or removal of the enclosure;
    - Quick and easy removal for inspection and maintenance is possible;
    - The enclosures do not impede the flow of cooling air over the fan motor when operating. Dedicated fans for ventilation of the motor within the acoustic enclosures may be required in some instances to prevent excessive heat build-up; and

- Be sufficiently robust to withstand a large number (not exceeding annual) of removal and replacement operations during the life of the fan, as well as be suitable for the installed environment.
- (vii) Dedicated nameplates manufactured from appropriate corrosion resistant material shall be affixed to each fan casing and motor by means of screws, engraved in accordance with the general specification.
- (viii) All items shall be individually labelled prior to packing and delivery. Labels shall include the following information as a minimum:
  - 1) Manufacturer name;
  - 2) Contract number; and
  - 3) Manufacturer's item/part number.
- (ix) Final agreement on exact label contents shall be in consultation with the S.O..
- (x) Where items are manufactured for specific mating component parts, they shall all bear individual identification numbers and reference to the mating part identification numbers.
- (xi) Liaison with the S.O. and timely provision of information is required to ensure installation can be completed in a timely and efficient manner.
- (xii) Protective Painting is required regardless of the fan guard material (including 316 Stainless Steel).
- (xiii) The Odorous Air Fan impeller shall be fabricated to withstand the stresses incurred during operation at the duty point as specified in the Particular Specifications for each site.
- (xiv) The thickness of the casing and impeller material shall be as specified by the manufacturer.
- (xv) Fan bearings shall satisfy the following requirements:
  - 1) Bearings shall be mounted out of the air stream;
  - 2) Bearings to have integral temperature monitoring instruments – Grade 316 Stainless Steel Resistance Temperature Detector (RTD) probes with loop powered 4-20 mA signal converters and connection junction boxes. The instruments and signal converters shall be intrinsically safe (e.g. "IA") and be certified to International Electrotechnical Commission (IEC) standards;
  - 3) Fan bearing housings shall be supplied with a clearly marked vibration monitoring stub on each bearing; and
  - 4) Fan bearings shall be of the type greased for life and designed for a rated fatigue life of one hundred thousand (100,000) hours at the specified duty point.
- (xvi) Fan vibration mounts shall satisfy the following requirements:
  - 1) Vibration mounts suitable for attachment to a concrete plinth shall be provided as part of the works by the Contractor. The concrete plinth for fan installation shall be excluded from the Contractor's scope of supply;

- 2) The vibration isolators shall be rated to dampen the vibration frequencies that may occur during the fans' operating speed range;
  - 3) The fan and motor shall be dynamically balanced and installed on bases designed to accommodate vibration isolators; and
  - 4) The fan and motor shall contain mountings selected to achieve ninety five percent (95 %) of vibration isolation efficiency at the normal operating speeds of the equipment.
- (xvii) Fan Acoustic Enclosures, if required, shall be selected based on its noise reduction performance, maintainability and its ability to withstand the elements.
- 1) For major overhaul, the enclosure shall be designed to allow for partial disassembly to allow access to both fans and motor by a lifting device. The enclosure shall be fitted with lifting points for complete removal;
  - 2) The enclosure shall be provided with a suitable method of ventilation for fan cooling. If ventilation fans are required, duty / standby ventilation fans shall be provided inside the acoustic enclosure to direct all exhaust air to outside of the building. The fans shall be temperature controlled with temperature sensors designed to signal an alarm in the event of the ventilation fan failing or over temperature of the odour fans;
  - 3) Sufficient small power and lighting shall be provided inside the enclosure and subject to S.O.'s approval;
  - 4) If filters are specified for fans, the filter pressure, loss indicator, and discharge pressure gauge shall be visible from outside the enclosure;
  - 5) Unless otherwise specified, the enclosure shall incorporate hinged doors or lift-off panels to provide safe access to all fan components requiring inspection, cleaning or maintenance. The structural integrity of the enclosure shall be maintained when one or more panels are removed. Doors/panels shall comply with the following requirements, as a minimum:
    - Doors shall open to a minimum angle of 90 degrees and be restrained against wind loads when open;
    - Panels shall be easily removed and manipulated by one person (e.g. fitted with suitable handles) and have a maximum weight of 25 kg;
    - Doors/panels shall be securely held in place when closed/in position and shall be lockable by padlock; and
    - Gaskets or seals shall be provided around doors/panels to maintain the acoustic performance of the enclosure.
  - 6) Where specified, 'walk-in' type acoustic enclosures shall be provided with all appropriate warning signs/notices;



- 7) The enclosure shall consist of an outer skin and a perforated inner skin. The space between the outer and inner skins shall be filled, unless otherwise specified. The material requirements are shown in G4.72.8 (c);
  - 8) All routine inspection and servicing shall be achievable without the need to remove the enclosure. Hinged access doors shall open to at least 90 degrees and be fitted with stay put type stays. Lift off access doors shall not exceed 25 kg in weight; and
  - 9) Enclosure discharge ventilation and pressure relief valve ventilation shall be ducted or vented away from the enclosure so that the high temperature air is dissipated away from the enclosure and any personnel access.
- (c) Exhaust Stack
- (i) The Contractor shall provide a self-supporting exhaust stack, if required, to achieve the design requirements as follows as a minimum, and incorporates the requirements as outlined in the Particular Specifications:
    - 1) Fabricated from vinyl ester FRP manufactured in accordance with BS EN 13121-3 and BS EN 13923 using chopped strand mat and woven roving backing a layer of surface tissue reinforced internal chemical barrier of minimum 2.5 mm thickness with C or ECR glass surface veiling, impregnated with vinyl ester resin. No metal exhaust stack is allowable;
    - 2) Suitable for operation under highly humid and potentially corrosive conditions as expected for sewer gas containing H<sub>2</sub>S, VOCs and other constituents typically found in sewer gas;
    - 3) Unless otherwise specified, the exhaust stack diameter shall be sized to accommodate an air velocity of approximately 15-20 m/s and up to 25 m/s; and
    - 4) Unless otherwise agreed with the S.O., the exhaust stack shall terminate at a minimum height of one building height above the tallest neighbouring building, to minimise downwash effects.
  - (ii) The Contractor shall supply the Exhaust Stack with a tundish and a drain line with “U” or “S”-bend water seal to allow for drainage of any accumulated condensate to the sump. The drain line shall satisfy the requirements outlined in Section G4.72.8(b)(iii).
  - (iii) The Contractor shall also supply the Exhaust Stack with appropriate sampling points (piped to ground level where required for accessibility), to satisfy the local regulatory authority’s requirements for air contaminant sampling.
  - (iv) The Contractor shall provide the Exhaust Stack with appropriate bolts for fastening onto a concrete footing.
  - (v) The Contractor shall design the Exhaust Stack for adequate stiffness and adequate footing size to limit deflections to a level that shall have no adverse impact on the foundation bolts or to any equipment or ducting connected to the stack.

- (vi) The external surface of the exhaust stack shall be painted with UV protection and a colour approved by The Board.

(d) Pre-Filter

- (i) A pre-filter shall be installed within the air intake duct upstream of odour fans for removal of aerosols, oils, grease, dust, and other airborne particulates from the incoming airflow to prevent blockage of the odour treatment media for the following conditions:

- 1) Where the air is extracted from dusty areas;
- 2) Where air is extracted from areas dealing with oils, fats, and greases;
- 3) Where the air is to be treated through an OCS utilising only AC; or
- 4) Where the air is to be treated through an OCS utilising only Biofilters.

- (ii) The pre-filter shall:

- 1) Be supplied with a self-contained sealed and lockable housing;
- 2) Remove greater than 99 % of aerosols of 4 µm or larger diameter at the design air flow;
- 3) Have a maximum face velocity across the filter of less than 2.5 m/s at the design maximum flow rate with all duty units online;
- 4) Be designed for continuous operation in odorous air which can be water saturated and contain corrosive/acidic gasses;
- 5) Be designed for the specified airflow rate as shown in the Particular Specifications;
- 6) The pre-filter tray/frame shall be designed for ease of disassembly, removal, inspection, and replacement without requiring the use of specialised lifting equipment;
- 7) Have pre-filter pads of standard replaceable size, the specific size requirements are outlined in the Particular Specifications for each site; and
- 8) Removable, disposable or washable panel filters are preferred. Each pre-filter pad shall be securely located within a non-corroding recessed tray/frame and held in position against a retaining mesh that is situated on the downstream side of the tray/frame.

- (iii) The pre-filter housing shall:

- 1) Have a hinged, air tight lockable access door for inspection, cleaning and replacement of filter pads;
- 2) Have flanged inlet and outlet for ductwork connections;
- 3) Have an isolation damper on the inlet and outlet sides of the housing so that the housing can be isolated from the odorous air stream;

- 4) Be provided with sample lines and a differential pressure gauge at accessible locations to monitor the pressure drop across the pre-filter (i.e. extent of blocking). The differential pressure gauge shall be clearly and permanently marked with the design values for “filters clean” and “filters dirty” conditions. The pre-filters shall not have greater than four hundred pascals (400 Pa) pressure drop across the entire pre-filter when fully blocked;
  - 5) Be provided with one condensate drain line to the sump fitted with a “U” or “S”-bend and water seal. Drain points for removal of condensate shall satisfy the requirements outlined in Section G4.72.8(b)(iii);
  - 6) Be provided with one (1) condensate drain line with DN25 ball valve for maintenance purposes, separate to the condensate drain line fitted with the “U” or “S”-bend and water seal;
  - 7) Have appropriate signage on access doors reminding to keep door closed; and
  - 8) Be constructed to withstand the full suction pressure of the fan (blocked duct conditions) so that permanent distortion of materials and breaking of seals does not occur.
- (iv) The Contractor shall provide details of the pre-filter and pre-filter housing in the returnable technical specifications table in the Particular Specifications.
- (e) Valves and Dampers
- (i) The Contractor shall provide valves and dampers for the isolation and control of air flow. Valves and dampers shall be designed to withstand ambient and process conditions with the presence of water and condensate along with highly corrosive and acidic gasses.
  - (ii) Valves and Dampers shall be free of rattles, fluttering or slack movement and capable of adjustment over the necessary range without excessive self-generated noise or the need for special tools. Valves and dampers constructed from mild or galvanised steel are not acceptable.
  - (iii) All valves and dampers shall be of heavy duty construction, double flanged, and designed to withstand up to 2-3 times the designed air flow and static pressure.
  - (iv) Where isolation of a process unit, filter or fan is required, and for isolation on major air headers and large dampers specifically intended for isolation or in a bypass duct, AMCA certified low leakage type butterfly valves shall be used to ensure effective isolation. Rubber seated tight shut butterfly dampers which meet AMCA Standard 511 for Class 1A leakage shall be used. Butterfly dampers shall provide positive sealing such that they have no leakage at 50 % higher than standard operating pressure. This shall be tested by isolating that section of duct and increasing the pressure in the duct to 50% above standard operating pressure for 15 minutes and measuring the loss in pressure to demonstrate greater than 99.5 % sealing.

- (v) The Contractor shall fit all extraction points with manually operated and lockable flow control dampers and isolation dampers and ensure the correct extraction flowrate is achieved during commissioning. All ducting going into or out of the odour scrubber vessels shall also be fitted with manually operated and lockable flow control and isolation dampers to ensure good flow distribution across all beds and vessels.
  - 1) The position of the flow control dampers shall be set at commissioning and locked in position for normal operation;
  - 2) The position of the flow control dampers may need to be re-set following maintenance and replacement of installed mechanical equipment, which includes but is not limited to maintenance and replacement of odorous air fans, maintenance and replacement of odour scrubbers or media; and
  - 3) Isolation of the relevant extraction point, or section of ducting/equipment will be achieved by manual adjustment of the isolation valve.
- (vi) Blades shall be without sharp edges and sufficiently rigid to eliminate movement when locked. The maximum clearance between blade and damper shall be less than one percent (1%) of duct diameter. Blades minimum thickness shall be as follows:
  - 1) Stainless steel = minimum 3.0mm
  - 2) FRP = minimum 10mm
- (vii) Damper bearings shall be oil impregnated sintered bronze ball bearings or engineering plastic sleeve bearings. Where the operating temperature is expected to exceed 50 °C, nylon shall not be used. Access for lubrication of bearings shall be provided.
- (viii) Spindles shall be mounted horizontally and securely fixed to damper blades with minimum diameter as follows:
  - 1)  $DN100 \leq \text{Damper diameter} \leq DN150$  = minimum diameter of 10mm
  - 2)  $DN200 \leq \text{Damper diameter} \leq DN400$  = minimum diameter of 16mm
  - 3)  $DN450 \leq \text{Damper diameter} \leq DN600$  = minimum diameter of 25 mm
  - 4)  $DN650 \leq \text{Damper diameter} \leq DN1300$  = minimum diameter of 30 mm
  - 5)  $\text{Damper diameter} \geq DN1350$  = minimum diameter of 35 mm
- (ix) The damper shall be capable of being adjusted and locked in the following blade positions: "Open", "10 ° to open", "20 ° to open", "30 ° to open" and "Closed". The positions shall be clearly and permanently labelled.
- (x) All non-return dampers assemblies, if supplied, shall be counterweighted so that it:
  - 1) Offers minimum resistance to air flow, and
  - 2) Closes by gravity.

- (xi) For isolating duty and standby odour fans automated butterfly valves with guard valves shall be provided in lieu of non-return dampers to ensure no backflow occurs through the standby fan.
- (xii) The requirements for dampers and valve actuation shall be as specified on the provided P&IDs and Drawings, and shall satisfy the requirements outlined in G4.1 Valves, Penstocks, and Stoplogs.
- (xiii) The acceptable materials of construction are outlined below:
  - 1) Body = Premium vinyl ester FRP
  - 2) Disc = Reinforced vinyl ester FRP
  - 3) Spindle = 316 stainless steel
  - 4) Bearings = Graphite filled Teflon or similar low friction material
  - 5) Handle and locking quadrant = 316 stainless steel
  - 6) Seals shaft material = EPDM
  - 7) Seal between disc and body material = EPDM or Viton
  - 8) Fasteners = 316 stainless steel
- (f) Access Ports
  - (i) Access ports shall be gasketed, sealed, and provided with facilities to prevent air ingress when not in use and shall be located such that monitoring instruments are easy accessible or visible from ground level.
  - (ii) Unless specified otherwise, ductwork shall be earthed to prevent build-up of static electricity and have a maximum resistance to earth 10  $\Omega$ , in accordance with PD CLC/TR 5040 (Electrostatics code of practices for the avoidance of hazards due to static electricity). Any equipment operating procedures or precautions required to prevent build-up of static electricity shall be listed in the tender submission.
  - (iii) Resilient seals and gaskets shall be fitted between all joints in the housing and ductwork to prevent air leakage.
  - (iv) Where non-return flaps are required to protect the unit from reverse flow, they shall be seated on resilient seals, thereby providing an effective seal against the reverse flow of air. The flaps shall be designed to offer minimum resistance to normal flow.
- (g) Access Hatches/Manways
  - (i) Unless otherwise specified, access hatches/manways shall have the following minimum clear opening:
    - 1) Personnel access: 900 x 900 mm.
    - 2) Hand access: 200 x 300 mm.
  - (ii) Access hatches/manways on ducting shall be double thick, deep formed, constructed and insulated to match the duct. Cold bridging shall be minimised. Panel frames shall be rigid, securely attached to the duct, with no part of the panel or frame protruding into the airstream.

- (iii) The seals on all access hatches/manways shall be mechanically fixed to either the panel or the frame to ensure an airtight seal when latched in the closed position. For fire rated seals, use woven ceramic fibre material. Latches shall be wedge type sash latches and there shall be a minimum of two.
- (iv) All access hatches shall be flanged, hinged, gasketed, sealed, fitted with transparent viewing ports, and constructed of the same material as the duct / vessel. The following requirements shall also be adhered to:
  - 1) Manways physically located on any odour scrubber media beds shall be constructed of same material as the vessel.
  - 2) Manways located in other locations shall be constructed as the same material as the duct/ vessel with a blank cover.
- (v) Sealed access hatches/manways shall be provided for each OCS treatment vessel at the following locations as a minimum:
  - 1) Bottom of the vessel, at the following locations:
    - Leachate sump;
    - Inlet plenum;
    - Under media support structures;
  - 2) Top of media, at the following locations:
    - Above each media bed/cell;
    - On top of the mist eliminator, if provided;
    - At spray assemblies (including emergency sprinklers), if provided;
    - At the outlet plenum, if provided.
  - 3) Top of vessel, to facilitate media replacement;
  - 4) At locations where instrumentation are installed inside the vessel; and
  - 5) At strategic locations to allow inspection and repair of the FRP duct/vessel internals.
- (vi) Means shall be provided for safely lifting the access hatches/manway covers into place/removal. All access hatches/manway covers shall be fitted with two (2) Nos. of lifting lugs and a davit shall be provided if the cover is greater than 16 kg weight. All lifting lugs shall be tested/certified and stamped with their working load limit (WLL).
- (h) Extra Tapping's
  - (i) The following extra tapping's shall be provided by the Contractor for the OCS:
    - 1) Manometer tapping's of minimum DN15 are to be positioned on the OCS inlet air odour duct for the measurement of air pressure, airflow, as well as for H<sub>2</sub>S and air quality sampling. The Contractor shall ensure the tapping points are piped to ground level such that they are accessible for measurement;

- 2) Where ducting or drain lines contain “U” or “S”-bends where water and material may accumulate, tapping’s of minimum DN25 are to be positioned on each “U” or “S” bend, with a drain line from the tapping to the sump, to allow inspection, top-up, and flushing of U bends;
- 3) Tapping on the exhaust stack for air quality sampling purposes shall be ninety degrees (90°) to each other with lengths being 75 mm each, unless otherwise specified in the particular specifications. The Contractor shall ensure the tapping points are piped to ground level such that it is accessible for measurement without the need for specialist or temporary access equipment;
- 4) Tapping on ducting for odour sampling purposes and air testing shall be of minimum DN15 and shall be in accordance with the sampling requirements as outlined in the specifications; and
- 5) All tappings shall be fitted with a removable seal/cap where required and fitted with gaskets or seals to prevent air leakage. Further details of are provided in the P&ID and general arrangement drawings supplied with this document Electrical, Instrumentation and Controls.

(i) General

- (i) Electrical equipment and instrumentation located within the odour control facility shall be certified as suitable for use in hazardous zones as classified in the Hazardous Area Classification;
- (ii) Local control stations shall comply with General Specifications G5 Electrical and G6 ICA;
- (iii) Electric motors shall comply with General Specification G5 Electrical; and
- (iv) Instrumentation shall comply with General Specification G6 ICA.

(j) Instrumentation

(i) Hydrogen Sulphide (H<sub>2</sub>S) analysers

- 1) H<sub>2</sub>S analysers shall be supplied to provide real-time verification of the OCS performance.
- 2) H<sub>2</sub>S analysers shall be installed at the following locations as a minimum, along with the local operator interface:

| Location                                                                                                                                         | Measurement Range (ppm) | Measurement Detection Limit (ppm) | Minimum No. of Readings per hour (No/hr) |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------|------------------------------------------|
| <b>OCF Treating High H<sub>2</sub>S and Odour Loads<br/>(Typically combined BTF and AC systems, Chemical Scrubber only, or BTF only systems)</b> |                         |                                   |                                          |
| OCF Common Inlet Duct                                                                                                                            | 0 – 200                 | 1                                 | Continuous on line, 1 minute averaging   |
| OCF Common Inlet Duct (only if inlet concentration range exceeds 200ppm)                                                                         | 0 - 1000                | 1                                 | Continuous online. 1 minute averaging    |
| Common Outlet Duct of first treatment stage                                                                                                      | 0 – 25                  | 1                                 | Continuous on line, 1 minute averaging   |

| Location                                                                               | Measurement Range (ppm) | Measurement Detection Limit (ppm) | Minimum No. of Readings per hour (No/hr) |
|----------------------------------------------------------------------------------------|-------------------------|-----------------------------------|------------------------------------------|
| Common Outlet Duct of first treatment stage (if two stages of treatment provided)      | 0 – 1.5                 | 0.1                               | 60                                       |
| OCF Common Outlet (i.e. duct upstream of Discharge Stack, or at surface of Biofilters) | 0 – 1.5                 | 0.01                              | 60                                       |
| <b>OCF Treating Low H<sub>2</sub>S and Odour Load Streams (Typically AC only)</b>      |                         |                                   |                                          |
| OCF Common Inlet Duct                                                                  | 0 – 10                  | 1                                 | Continuous on line, 1 minute averaging   |
| OCF Common Inlet Duct                                                                  | 0 – 1.5                 | 0.01                              | 60                                       |
| OCF Common Outlet Duct                                                                 | 0 – 1.5                 | 0.01                              | 60                                       |

3) Each hydrogen sulphide analyser shall require the following in the panel:

- Moisture knock-out pot;
- Naflon filter;
- Rotameters with needle valves for airflow adjustment;
- Suction pump;
- Mass flow switches for each pump; and
- Grade 316 stainless steel tubing or Teflon tubing inside the panel.

4) The stainless-steel cabinet shall be UV-resistant and rated IP 56. Any transparent sections shall be tinted to prevent analyser overheating or damage to analyser displays. The cabinet shall come complete with a light, lockable door and air conditioning (rated to IP 56) as required by the instrumentation it is to house.

5) Analysers shall come with a conditioning unit to reduce gas humidity and incoming grease to suit the requirements of the analyser.

6) Sampled gas discharge pipework shall be discharged at a location external to any instrument cabinets or enclosures. Sampled gas shall be discharged to the odour duct prior to the OCS inlet and not to atmosphere. The Contractor shall ensure that the sampling location and sampled gas discharge location considers other sample lines to prevent contamination of samples.

(ii) Flowmeters

1) An orifice plate type flowmeter shall be provided for measurement of airflow through the odour treatment system. Alternative types of flowmeters will not be accepted without prior express approval from the S.O..



- 2) The flow meter shall be installed complete with all necessary accessories and ancillaries required to form a complete and functional system. In particular, the flowmeter must satisfy the minimum straight lengths stated in the table below to ensure adequate accuracy of installation.

| Parameter                          | Units | Value             |
|------------------------------------|-------|-------------------|
| Minimum Upstream Straight Length   | -     | 5 x Pipe Diameter |
| Minimum Downstream Straight Length | -     | 3 x Pipe Diameter |

- 3) The installation shall incorporate a suitably located tapping point to allow calibration of the flow meter using another calibrated portable airflow meter.
- 4) The differential pressure gauge used to measure the pressure differential across the orifice plate shall be piped to ground level, for ease of access for maintenance.

(k) Control Narratives

- (i) A detailed Functional Design Specification (FDS) shall be prepared by the OCS Manufacturer and submitted for approval by the S.O. Refer also to Particular Specifications P6 ICA.
- (ii) The Contractor shall advise the alarm and interlock settings for the safe and reliable operation of the OCS and include these in the FDS.
- (iii) Emergency fire-fighting sprinkler systems, if provided (such as for AC Scrubber Vessels), shall be designed such that the system is fully automated via its own integral control system and will operate when high temperature is detected within the vessel.
- (iv) The Control Narrative shall include but not be limited to the following:
  - 1) Controlled starts and stops;
  - 2) Any signals required from upstream equipment;
  - 3) Any safety interlocks required for personnel or equipment protection;
  - 4) Interlocks and alarms required to meet National Environment Agency (NEA) conditions as outlined in the Particular Specifications and includes the following as a minimum:
    - High concentrations of H<sub>2</sub>S in discharge air; and
    - High flow alarm.
  - 5) Required equipment failsafe state in the event of power failure;
  - 6) Emergency stop requirements; and
  - 7) Alarm requirements, with the following as a minimum:
    - Low flow alarm from the odorous air fans;
    - High H<sub>2</sub>S concentrations in the treated air streams;

- High temperature alarm from the AC media beds;
- High CO reading alarm from the AC vessel exhaust; and
- High-high temperature alarm from the AC media beds.

#### **G4.72.14 Scrubber Vessel Design Requirements**

##### **(a) Overview**

- (i) The performance specifications provided below are for the information of the Contractor. They give general loadings, load combinations and guidelines of features that the Contractor shall incorporate into the final design. Any changes to these features that the Contractor may find necessary in his design work must be approved by the S.O..
- (ii) The information listed below shall be read in conjunction with the General Specifications G4.60 Pressure Vessels and G4.61 FRP and Thermoplastic Tanks. Proposed odour scrubbers must satisfy all requirements in G4.61 and G4.61 in addition to the information presented in this document. Where there is a conflict between the information presented in G4.60 or G4.61 and the information presented herein, the more stringent criteria shall take precedence.

##### **(b) Design Inputs**

- (i) All odour scrubbers shall be a self-contained and sealed, counter-current flow, free standing tower anchored to concrete slabs on the ground and fitted with all necessary internals to provide a fully functional system. The Contractor shall be required to complete the structural and mechanical design of each of the scrubber vessels, internal support beams (or equivalent), internal media support plates, internal turning vanes and internal distribution pipework (as indicated on the drawings).
- (ii) The Contractor shall ensure that all design work is undertaken by a Professional Engineer (P.E.) from the appropriate discipline. The Contractor shall make allowance for additional ductwork loading supported by the vessels. No cross-flow scrubbers are permissible.
- (iii) Further details on design inputs for the odour scrubber vessels can be found in the Particular Specifications.
- (iv) The design shall minimise the gas side differential pressure across the tower.
- (v) Unless otherwise specified in the Particular Specifications, the vessel diameter shall not exceed 4.0 m. For proposed vessel diameters requiring special delivery considerations for ground transport, the contractor shall outline the proposed methods to deliver the vessels to the site, and ensure that any associated costs, such as police escorts, and the use of special routes, are included in the proposed tender price.

- (vi) Unless otherwise specified in this Specification or in the Particular Specifications, all nozzles on odour scrubbers shall be flanged with a minimum nozzle size of 50 mm Nominal Diameter (ND). They shall be located for easy access and to minimise quantity of piping. Nozzle locations shall match those on attached drawings. Gaskets shall be provided for nozzles and be a minimum 3 mm thick Ethyl Propylene Rubber (EPR) full face gaskets. Press-moulded flanges are not acceptable.
- (vii) No exposed metal items shall come in contact with any process gas and liquid streams.
- (viii) Certified lifting lugs shall be designed, certified and provided on each vessel segment and any equipment requiring crane assembly/disassembly for maintenance or repair. All lifting lugs shall be stamped with their certified Working Load Limit (WLL).
- (ix) All inlets and discharge ducting connections on each odour scrubber vessel shall be flanged.
- (x) The inlet plenum shall be designed to ensure even airflow distribution through the odour scrubber media. Each odour scrubber inlet plenum shall be provided with a dedicated flow balancing damper and an Air Movement and Control Association (AMCA) certified low leakage isolation damper. Each scrubber outlet plenum shall be provided with an AMCA certified low leakage isolation damper for isolation.
- (xi) Each scrubber vessel outlet duct shall have a design air exit velocity of less than 10 m/s for the specified operating airflow as specified in the Particular Specifications with all units in service (i.e. normal operating conditions).
- (xii) Each odour scrubber vessel shall be fitted with a differential pressure indicator transmitter fitted across each bed to assist in identifying blockages and determining if replacement of media is required due to media degradation.
- (xiii) Each odour scrubber vessel shall be installed and mounted such that it is not necessary to disassemble the vessel for access to remove and replace the media in each bed within the vessel.
- (xiv) The odour scrubber vessel shall be designed to support the packing media selected by the Contractor, plus entrained humidity and/or biomass/biofilm growth on the media.
- (xv) The finished vessel wall, internally and externally, shall be free of defects such as foreign inclusions, air bubbles, pinholes, pimples, crazing, cracking, and delamination that may impair the serviceability of the vessel.
- (c) Vessel Pressure
  - (i) The design, operating, and test pressures of the vessels shall satisfy the requirements as outlined below or otherwise specified in the Particular Specifications.
    - 1) Design pressure of each vessel shall be greater than or equal to 2 times the operating pressure; and

- 2) The test pressure of each vessel shall be greater than or equal to 1.3 times the design pressure in accordance with BS EN 13121-3 and BS EN 13923.

(d) Mechanical Design Requirements

(i) Vessel Attachments

- 1) Provide pipe supports integral to tank for any down pipes or pipes supported from the vessel walls;
- 2) Pipes shall be supported at maximum 1,500 mm intervals or more frequently as may be required by the relevant Standard; and
- 3) Details of pipe supports shall be confirmed with the S.O. prior to installation.

(ii) Media Support

- 1) Quantity: Sufficient to hold odour scrubber media plus entrained irrigation liquid and/or humidity and/or biomass/biofilm growth on the media;
- 2) Specific media support requirements for odour scrubbers:
  - The media support directly underneath AC media shall where possible contain a minimum of 200 mm layer of inert large diameter material (e.g. clay balls, pebbles, or plastic packing) for any potential leachate to be collected and drained;
  - Opening size of media supports shall not allow odour scrubber media to fall through; and
  - Media support shall be structurally supported to limit the mid-span deflection to  $(L/125)$  where  $(L)$  is equal to the span length in mm.

(iii) Sampling Ports

- 1) Each odour scrubber vessel shall be fitted with H<sub>2</sub>S logging and air quality sampling points in the inlet and outlet plenums (i.e. areas of low air velocity of less than 0.25 m/s) with the following requirements:
  - The sample points shall take the form of access hatches which allow access to the inlet and outlet plenums;
  - The inlet and outlet plenums are to be fitted with hanging hooks for H<sub>2</sub>S data logging units which can be installed and reached from outside the vessel without the need for personnel to insert their heads into the access hatches to observe or handle the H<sub>2</sub>S data logging units; and
  - The access hatches are to have suitable sealing gaskets bonded to the access hatch and be able to be locked.

- (iv) Vessel Access Platforms
  - 1) The Contractor shall provide each odour scrubber vessels with access platforms and stairs to allow safe and convenient access for operations and maintenance activities (e.g. calibration, adjustment, media replacement, etc.) and for the inspection of internals (e.g. fire-fighting nozzles and Bio Trickling Filter irrigation spray pattern, instrumentation, media conditions, etc.) via the inspection ports. Access platforms to inlet and outlet air isolation dampers shall be provided.
  - 2) It shall comply with relevant codes, including relevant OH&S Acts, Regulations and Codes of Practices.
  - 3) The access platforms shall be provided complete with access stairs and have a kick plate around its perimeter and other design features that will facilitate safe operation such as during change out of scrubber media.
- (e) Design Loadings
  - (i) Load Combinations
    - 1) Design for the combination of loading shall be in accordance with the General Specification G3 on Civil and Building Works.
- (f) FRP Vessel Requirements
  - (i) All odour scrubbers, including all internal support beams (or equivalent), internal media support plates, turning vanes and distribution pipework shall be manufactured of premium corrosion resistant vinyl ester FRP manufactured in accordance with BS EN 13121-3 and BS EN 13923 using chopped strand mat and woven roving backing a layer surface tissue with reinforced internal chemical barrier impregnated with vinyl ester resin, suitable for long term use under highly corrosive conditions.
  - (ii) Materials of construction - The resin used must:
    - 1) Be suitable for the intended service;
    - 2) Premium grade and corrosion resistant type, such as Bisphenol A Vinyl ester (e.g. Hetron 922, Derakane 411-350, Swancor 901, Dion 9100 Epovia RF1001). Any other resins are not acceptable without prior express written approval by the S.O.;
    - 3) Heat Distortion Temperature (HDT) of resin shall be greater than 20 °C over the design temperature;
    - 4) No fillers or thixotropic agents are permitted in the internal corrosion layers;
    - 5) No dyes, pigments, or colorants are permitted, except in exterior gel coat. The vessel and ducting colour will be confirmed; however, the Manufacturer shall be prepared to supply the vessel in any colour as determined by The Board; and
    - 6) Ultraviolet absorbers shall be added to surfacing resin to improve weather resistance.

(iii) Corrosion allowance:

- 1) The inner surface exposed to the contents shall be a resin rich layer reinforced with a surfacing veil providing a thickness of 0.25 mm to 0.5 mm. The inner layer corrosion resistant barrier shall be designed as per BS EN 13121-1;
- 2) The inner surface layer, exposed to the contents, shall be followed with an interior layer. Construct this layer using resin reinforced with non-continuous glass fibre strands (25 mm to 50 mm long), applied in a minimum of two plies totalling a minimum of 900 g/m<sup>2</sup>. Piles are to be made from layers of either chopped strand mat or chopped roving or both to achieve the required strength. Each ply of mat or pass of chopped roving shall be well rolled to thoroughly wet out reinforcement and remove entrapped air prior to the application of additional reinforcement. The combined thickness of the inner surface and interior layer shall not be less than 2.0 mm; and
- 3) The reinforcement content of the corrosion barrier shall be 25 % to 35 % by weight.

(iv) The structural layer shall:

- 1) The structural layer shall be constructed using filament wound or contact moulding methods;
- 2) Application of the structural layer shall not alter the corrosion-resistant barrier;
- 3) The first ply of the structural layer of the laminate shall be one or more plies of non-continuous glass fibres totalling 300 g/m<sup>2</sup>, comprised of chopped strand mat and/or chopped roving;
- 4) Continue lay-up in the sequence of plies stated for the specified laminate type, noted in BS EN 13121-3;
- 5) All edges of reinforcement material shall be lapped a minimum of 25 mm. Lapped edges of adjacent layers shall be staggered; and
- 6) The structural layer materials shall comply with the guidelines set out in BS EN 13121-3.

(v) The outer surface layer shall:

- 1) Meet visual acceptance criteria as in BS EN 13121-3. in addition to those included within this document;
- 2) The outer surface of the finished laminate shall be a separately applied paraffinated resin coat that, when cured, passes the acetone test;
- 3) The laminate surface shall be provided with protection from UV degradation by use of an ultraviolet absorber incorporated into the outer surface layer. Any use of pigmentation in the outer layer shall only take place after visual inspection of the vessel by the S.O.; and

- 4) Where the final lay-up is exposed to air, full surface cure shall be obtained by applying to the final layup a coat of paraffinated resin that, when cured, passes the acetone test. Other techniques such as sprayed, wrapped, or overlaid films are also acceptable methods to attain surface cure, provided the surface resin under the film passes the acetone test.
- (vi) The curing system shall be:
- 1) As recommended by the resin manufacturer or as specified;
  - 2) Tested through measurement of the Barcol hardness according to ASTM 3572.22:1997. The resin must display Barcol hardness of 90 % or more of the manufacturers prescribed resin hardness; and
  - 3) Post-cure vessels, ducting and appurtenances in accordance with the resin manufacturer's recommendation for time and temperature. Post-curing should be completed with warm-to-hot dry air, free from combustion products. Localised hot spots shall be avoided.
- (vii) Reinforcement taking place through, for example, local thickening will be required in areas subject to concentrated loads or subjected to wear, these include:
- 1) Hold downs;
  - 2) Anchor bolts;
  - 3) Brackets;
  - 4) Supports;
  - 5) Lifting points; and
  - 6) Other areas that experience significant stresses or wear that will shorten the life of the product.
- (viii) Flanges:
- 1) Flanges should be flat-faced design for use with backing flanges. They must allow for washers to be used under nuts and cut outs e.g. those required for bolt insertion. The edges must be sealed with resin to enhance durability. For pipe flanges, design them to meet the requirements of AS2129, AS2834-1983 or ANSI B16.5 #150. For other flanges design flange thickness to prevent flange distortion and leakage between sealing faces.
- (ix) Pipe or vessel connections:
- 1) Shall be of butt and strap or adhesive joint type, however preference will be given to butt and strap joints. Joints must be designed to the guidelines set out in BS EN 13121-3 and BS EN 13923; and
  - 2) Branches shall be as short as possible but of sufficient length to permit removals of bolts from flanges and to permit the required overlay length for lap shear. All branch pipes shall be designed as per BS EN 13121-3.

- (x) The following guidelines and manufacturing tolerances must be followed:
  - 1) Minimum wall thickness shall be provided to withstand the loading due to hydrostatic forces, wind loads and other associated loading to be determined in accordance with BS EN 13121-3. Alternative standards may be used subject to written approval by the Purchaser; and
  - 2) Tolerances shall be generally in accordance with BS ES 13131.
- (xi) Openings, Branches and Compensation
  - 1) Openings
    - The requirements of openings shall be as per BS EN 13121-3 and BS EN 13923. The openings in scrubber vessels shall be circular or oval with a ratio of major to minor axis not exceeding 2. All openings shall be compensated using additional laminate designed in accordance for all planes through the axis of the openings normal to the vessel surface;
    - All nuts and bolts to be provided and installed using nickel anti seize, suitable for the application under highly corrosive conditions.
  - 2) Bolted Flanged Circular Connections
    - The requirements of bolted flanged circular connections shall be in accordance with the drawings.
  - 3) Stiffening Rings.
    - The requirements of stiffening rings shall be determined as per BS EN 13121-3 and BS EN 13923 and shall extend completely around the circumference of the shell and any joints shall be so designed as to develop the full stiffness of the ring. There shall be no spaces between the ring and the shell. The design shall be checked for the stresses in the FRP caused by restraint to thermal expansion/contraction by stiffening rings where the design temperature is above 60 °C or below 0 °C.
- (xii) Odour Scrubber Vessel Anchorage
  - 1) The Contractor shall be responsible for the design and supply of vessel anchor bolts. Sizing of required vessel anchor bolts shall be undertaken by a P.E. of the relevant discipline. The requirements for vessel anchor bolts shall be as per BS EN 13121-3 and BS EN 13923; and
  - 2) The number of anchorage points shall be a multiple of four (4), be evenly spaced around the diameter, and the spacing should not exceed 1,500 mm. The anchor bolts shall be designed and arranged to allow for the bottom branches to be sized so that there is room for gussets and so that high local loads are not imposed on the additional reinforcement which is used for compensation of the branch opening.



- (xiii) Odour Scrubber Vessel Support
  - 1) All supports shall be designed by a P.E. of the relevant discipline and shall accommodate temperature changes and permit free expansion and contraction. Where there is steel bonded into the support, the design shall be checked for stresses in the FRP caused by restraint to thermal expansion/contraction where the design temperature is above 60 °C or below 0 °C; and
  - 2) The proposals for support shall be agreed with the S.O. at the design stage and the agreed details shall be specified on the approved drawings. The requirements for vessel support shall be as per BS EN 13121-3 and BS EN 13923.
- (xiv) Other Structures and Fittings
  - 1) All structures and fittings shall be arranged as far as practicable to avoid imposing local concentrated loads on the walls of the scrubber vessels. The design shall be such that the local strain does not exceed the allowable strain as per BS EN 13121-3 and BS EN 13923. The proposals for detailed design of all such structures and fittings shall be agreed with the S.O. at the design stage and the agreed details shall be specified on the approved drawings; and
  - 2) Suitable earthing shall be provided for all scrubber vessels.

#### **G4.72.15 Product Delivery, Storage and Handling**

- (a) The equipment shall be delivered to site as stipulated in the Particular Specification. The cost of storage and delivery of said equipment shall be deemed included in the Contract Rates submitted by the Contractor.
- (b) The equipment shall not be delivered to site until the site is able to accept the equipment, and the S.O. has authorised delivery. The Contractor shall inspect the site prior to delivery and confirm that site works have been completed in accordance with their requirements and there will be no delay to the installation of their equipment when delivered to site.
- (c) The S.O. may elect to inspect the equipment to be supplied by the Contractor prior to packaging and delivery. The S.O. will give a minimum of five (5) business days' notice to the Contractor of an intended inspection.
- (d) All equipment delivered to site shall be packed to protect it against damage during loading, transport, unloading and storage.
- (e) Equipment stored onsite shall be sealed and protected and be suitable for outside storage. All exposed equipment openings shall be properly protected. Appropriate measures shall be taken to prevent the entrance of moisture or water to equipment during shipment and storage onsite for a period of up to NINETY days.
- (f) Biofilter media, if supplied, shall be adequately protected from contamination and protected from damage during delivery, storage, and handling.
  - (i) Support boards or FRP grating shall be provided such that if personnel are expected to step on the media, their weight can be supported without displacing or damaging the media.

- (ii) Care shall be taken to ensure that media layers are not compacted during temporary storage, placement, and handling.

#### **G4.72.16 Specific Testing and Commissioning Requirements**

- (a) General

- (i) Testing and commissioning requirements for the OCS shall satisfy the requirements outlined in G2.26 General Specifications for Plant Testing and Commissioning, in addition to the specific requirements outlined in this document. Where there is a discrepancy between the two documents, the more stringent requirement shall apply.

- (b) Testing Plans

- (i) Testing plans submitted for the OCS shall include, but not be limited to, testing dates, durations, pressures, vessel orientation, blanking plates and areas to be blanked and testing location.

- (c) Factory Test (FAT)

- (i) FRP Vessel Testing:

- 1) Prior to any factory testing, the corrosion barrier shall be properly cured and hardened in accordance with BS EN 13121-3 and BS EN 13923. No part of the vessel may be flow coated prior to testing.
    - 2) All equipment shall be inspected as per the guidelines set out in BS EN 13121-3. Inspection and Testing shall be completed by an independent GRP/FRP inspector. Visual inspection shall take place following the guidelines listed in Table 24 of BS EN 13121.
    - 3) The following quality and testing requirements are valid for all GRP/FRP components as part of these works:
      - During manufacture a Manufacturer appointed representative may perform a number of inspections at the Manufacturer's works;
      - Manufacturer's Quality Assurance Supervisor should have a minimum of FIVE (5) years' experience in fabrication of fibreglass structures;
      - The designer shall be a P.E. with at least five (5) years' experience in the design of fibre reinforced plastic structures and components. Designer qualifications shall be submitted with the Contactors proposal; and
      - The tolerances for wall thickness of components and vessels must comply with BS EN 13121-3.

- (ii) Required acceptance levels

- 1) Descriptions of defects and the criteria for acceptance levels shall be as per BS EN 13121.

(iii) Fan Factory Testing

- 1) The fans shall be factory tested as per BS EN ISO 5801 by the Contractor at 25 %, 50 %, 75 % and 100 % of the specified maximum duty points to verify that they are capable of handling the new duty points. Motor input power and inlet and discharge pressures will also be measured and recorded at the test points. The Contractor will provide fan curves for each fan based on these measurements;
- 2) Dye penetrant and magnetic particle tests of welding shall be carried out to the satisfaction of the S.O.;
- 3) Test results of impeller balancing tests shall be submitted by the Contractor prior to factory testing; and
- 4) Housings shall be tested in accordance with the relevant standard and where appropriate, test results provided.

(iv) Pressure Testing – Clarifications to BS EN 13121-3 and BS EN 13923 (Scrubber vessels only)

- 1) All scrubber vessels (both AC vessels and BTF vessels) shall be tested to BS EN 13121-3 and BS EN 13923 with the following amendments/additions;
  - All completed tanks and vessels manufactured in FRP to BS EN 13121-3 and BS EN 13923 Cat. 1 shall be subjected to a pressure test in the manufacturer's workshop. After installation on site the tank or vessel shall be re-tested by the Contractor; and
  - The manometer used to measure the test pressures shall indicate up to 50 % more than the maximum expected pressure.
- 2) The Contractor shall produce all necessary documentation and/or records, including third party inspections as detailed in BS EN 13121-3 and BS EN 13923 Cat.1 regulations, as verification that factory testing has been successfully completed. An up to date copy of developing test records shall be available always for information and review.
- 3) Testing of Scrubber Vessel Operation at Elevated Pressure
  - An optional pricing shall be included in the returnable schedule for the testing at elevated pressure;
  - The scrubber vessel shall be tested in the same orientation, as it will be in service;
  - The test pressure shall be determined from the design code and add an allowance for wind loading for vessels, which will be installed outdoors;
  - The pressure shall be increased to 50 % of the test pressure and held at this pressure for 15 minutes to test for joint leakage. If the pressure cannot be held then the pressure shall be released and all joints re-tightened and the test restarted;

- When the pressure holds for 15 minutes at 50 % of the test pressure then pressure shall be increased in 10 % of test pressure stages with a 5-minute hold at each pressure;
- When the test pressure is reached the vessel shall be isolated from the pump and the manometer monitored for 1 hour to confirm that there is no leakage;
- At the end of the hour the vessel shall be reduced to 90 % of the test pressure and the exterior inspected for damage and leaks by the 3<sup>rd</sup> Party Inspector;
- Once the 3<sup>rd</sup> Party Inspector has completed the inspection of the exterior and deemed a pass or fail result inspection the vessel can be vented; and

4) Pass/Fail Criteria

- Changes in internal pressure caused by fluctuations of ambient temperature/pressure (which shall be monitored and recorded) will not be accepted as reasons for failure of the test;
- Vessels are deemed to have passed the pressure test if they can hold the test pressure for a minimum of 1 hour in a sealed condition;
- Vessels are deemed to have failed the pressure test if they cannot hold the test pressure for a minimum of 1 hour in a sealed condition;
- Any distortion or change in shape of the vessel during the test will be reason for failure of the test; and
- Any failure of damage to the laminate will be reason for failure of the test.

(d) Site Acceptance Testing (SAT) / Pre-Commissioning Testing

(i) Dry Testing

- 1) Following installation of the equipment at site, any components/systems that have not been previously inspected during FAT or have been newly assembled/installed at site shall be subjected to a thorough visual inspection to confirm that they have been correctly assembled/installed in accordance with the relevant P&ID/general arrangement drawings.

(ii) Wet Testing (With Water or Final Effluent)

- 1) Prior to the commencement of wet testing, the equipment shall be flushed/cleaned to remove any debris present in the system;
- 2) Pipework sections and all associated connections shall be pressure tested at a pressure of 1.5 times the pressure relief valve setting or maximum system pressure, as appropriate, subject to a minimum pressure of 2.5 bar. There shall be no loss of pressure or visible leakage after a minimum of 1 hour and/or specific period stated in the specification;

- 3) The equipment shall be subjected to mechanical and electrical functional testing, over its full range of control settings to verify its satisfactory operation under site conditions. Where appropriate, the following components/systems shall be tested for correct operation, as a minimum:
    - BTF/Biofilter Irrigation systems;
    - Biofilter Humidification systems;
    - BTF irrigating liquid nutrient dosing systems;
    - BTF irrigating liquid re-circulating systems;
    - BTF irrigating liquid blowdown systems;
    - Pre-filters (if supplied, for AC only systems or Biofilter only systems)
    - Dehumidifiers/heaters (if supplied for AC based systems)
    - Odour extraction fans;
    - Monitoring instruments and protection system (e.g. pressure gauges/transducers, level monitors, flow meters, flow switches, pH and H<sub>2</sub>S analysers, etc.). All instrumentation to be calibrated as per manufacturers recommendations; and
    - All relevant alarms.
  - 4) If the equipment is to be left empty for a prolonged period, it shall be thoroughly dried by blowing/sucking dried air through it using a suitably sized compressor/vacuum pump; and
  - 5) Equipment storage areas shall be provided with dehumidification facilities to prevent condensation occurring within electrical control panels, if necessary. If dehumidification facilities are unavailable, desiccant shall be placed within individual compartments.
- (iii) Further Tests (Where Required)
- 1) Details of any further site acceptance tests to be performed on the treatment unit shall be as specified in the Particular Specifications and as per S.O.'s instruction without incurring any additional cost to The Board.
- (e) Site Performance Testing (SPT)
- (i) General
- 1) All equipment within the OCS package shall be installed in strict accordance with the Contractor's design details, shop fabrication drawings, and the original equipment manufacturers recommended instructions.
  - 2) The Contractor's instructions shall include any special requirements for mounting and installing of odour treatment vessels, scrubber media / treatment media, and any requirements for installation, balancing, alignment and levelling.

- 3) After all required equipment has been installed, acceptance checks will be performed by the Contractor to confirm acceptance of installation and construction completeness, with the S.O.'s representative in attendance.
- 4) All odour covers and ducting sections shall be smoke tested to ensure there are no leaks identified for a minimum of one (1) hour. Furthermore, an airflow balance test shall demonstrate that there is no more than 2.5 % leakage through the system from the source to point of discharge.
- 5) All pipework sections and all associated connections shall be pressure tested at a pressure of 1.5 times the maximum design pressure in accordance with BS EN 13121-3. There shall be no loss of pressure or leaks for a minimum of 1 hour.
- 6) The Contractor shall be responsible for airflow balancing for each treatment vessel/cell to ensure that equal distribution to each unit (the lesser of the Contractor's recommended tolerance or +/- 5 % of the design conditions).
- 7) The S.O. reserves the right to request its representatives be present for any testing.
- 8) Any equipment used for measuring H<sub>2</sub>S concentrations shall:
  - Be suitable for the environment in which it will be used (including hazardous areas, where appropriate);
  - Be capable of sampling, measuring and recording the required H<sub>2</sub>S concentrations with an acceptable level of accuracy, at the required intervals for the required time period;
  - Incorporate a power supply that permits the required monitoring regime to be achieved;
  - Incorporate a sample collection air pump, if necessary;
  - Have a current calibration certificate within the past six (6) months relevant to the range of H<sub>2</sub>S concentrations measure;
  - Be capable of measuring and recording zero drift errors;
  - A record of known cross-sensitivities to other gaseous compounds shall be documented (e.g. mercaptans);
  - Monitoring equipment used for measuring H<sub>2</sub>S concentration at the inlet of the treatment unit shall have an accuracy of +10 %, up to the maximum anticipated H<sub>2</sub>S concentrations;
  - Monitoring equipment used for measuring H<sub>2</sub>S concentration at the outlet of the treatment unit shall be capable of measuring H<sub>2</sub>S concentrations down to an appropriate level, depending upon the concentrations being measured; and

- The average and maximum H<sub>2</sub>S concentrations shall be calculated for each sampling/data logging session. The average and maximum H<sub>2</sub>S concentrations shall not exceed the value specified as in the Performance Guarantees outlined in the Particular Specification.

(ii) Fan Performance Testing

- 1) The motors shall be tested prior to assembly onto the new fan. The tests shall be carried out where relevant in accordance with relevant standards;
- 2) Basic (i.e. type test) certificate shall be issued to the S.O. by the Contractor in the tender proposal. Routine test certificates including individual load/efficiency curves for each motor shall be supplied, certified by the motor manufacturer. Certificates for these tests shall be provided in advance of any further efficiency tests on plant driven by the motor tested; and
- 3) Fan performance testing shall satisfy the following:
  - Each fan shall be given a site mechanical run test at the duty operating speed and full design load for one hour per operational speed using fresh air, for a combined duration of no less than 24 hours;
  - Each fan shall be tested with ventilation air and operated over its full operating range for the performance test. During the performance test, each Fan must run continuously and satisfactorily in automatic mode for a minimum of 48 hours, and the Fans must achieve a minimum of seven (7) days “trouble free” operation of all equipment (i.e. no major breakdown or modification);
  - During the mechanical run tests, the operation of every part shall be satisfactory and complete to the specified duty, noise and vibration requirements; and
  - If during the site mechanical testing runs the fan does not meet the requirements specified, then repairs are to be undertaken. The initial run test shall be considered void and shall be re-run. A complete detailed log of all final tests shall be assembled and submitted to the S.O. on completion of the tests including impeller balancing and vibration measurement and noise testing. Data should be taken over the operating speed range. An onsite balance test shall also be undertaken if required following on site Vibration tests.

(iii) Air Distribution Testing – for Biofilter Systems only

- 1) Smoke tests shall be performed on the installed Biofilter system (including media) to ensure uniform air distribution and to confirm no short circuiting is occurring at the Biofilters. Each smoke test of the Biofilter shall be a minimum of ten (10) minutes in length. Particular attention shall be given to the following areas:
  - Perimeter walls of the Biofilter.

- Joint between the duct and the air distribution plenum
  - Plenum joints and any biofilter cell access doors
  - Around the Biofilter cell drain penetration
- 2) During the smoke test, the relative flow from each Biofilter cell and the whole Biofilter installation shall be measured by placing a flux hood connected to a flowmeter, over a suitable central position of each installed duty Biofilter cell. Each Biofilter cell shall have a flowrate within five percent (10%) of the design flowrate per Biofilter cell.
- 3) For the use of soil bed media Biofilters, prior to commencing smoke testing, the moisture content and bed density of each Biofilter cell shall be measured using a portable moisture probe. Measurements shall be taken at depths of at least 0.5 meters, and at least ten evenly spaced locations per cell shall be measured, or as otherwise specified. This step is not required for engineered media Biofilters.
- 4) Following the first smoke test, the following measurements shall be recorded for eight (8) hours at the indicated frequencies:
- Hourly measurement of the total airflowrate to the Biofilter system, using a handheld anemometer;
  - Hourly measurement of the differential pressure at each Biofilter cell, using a handheld differential pressure sensor;
  - Two (2) measurements of airflow rates taken at four (4) locations for each Biofilter cell. One measurement shall be taken at the beginning of the eight (8) hours, and the other taken at the end of the eight (8) hours. The measured airflow rates at the beginning and end of the eight (8) hours shall not deviate by more than ten percent (10%) across each Biofilter cell. If this is not achieved, Biofilter media shall be removed, topped off, or replaced by the Contractor, as required by the S.O., to achieve proper airflow distribution.
- 5) A second smoke test shall be conducted after irrigating the Biofilter cells for one (1) irrigation cycle at the irrigation cycle length stipulated by the Manufacturer / Contractor. The plenum pressure and air flowrate shall be recorded during this second smoke test.
- 6) Following each smoke test, any airflow imbalances identified shall be rectified.
- Rectification works shall include but are not limited to removing, topping off, or replacing Biofilter media as directed by the S.O. to achieve the proper airflow distribution required.
  - Following any modifications to the Biofilter system as indicated by the smoke test findings, the Biofilters shall be smoke tested once more to demonstrate that proper airflow distribution is achieved.



- Rectification works to the Biofilter and further smoke testing to verify proper airflow distribution shall be repeated until the results are considered satisfactory to the S.O., and the S.O. is satisfied that proper airflow distribution has been achieved.
- 7) The Biofilters will only be accepted by the S.O. upon successful smoke testing, as evidenced by an even rise of smoke over the entire Biofilter bed, with no leakage points. All testing shall be conducted in the presence of the S.O., and a minimum of two (2) weeks' notice shall be provided to the S.O. prior to testing.
- (iv) Humidification and irrigation spray testing – for Biofilter Systems only
  - 1) The Biofilter humidification and irrigation system shall be flow tested to verify that each irrigation zone and humidification chamber performs at the specified flowrates at the specified pressures. The duration of the test shall be at least one (1) hour.
  - 2) Prior to the humidification and irrigation spray testing, the Biofilter shall be operated for a minimum of twenty-four (24) hours in the AUTO mode in the PLC and SCADA. During this period, the Contractor shall monitor the Biofilter humidification and irrigation system operation to ensure that the correct flowrates and the correct pressures are being delivered. Adjustments should be done as necessary to deliver the correct flowrates and pressures to the humidification chambers and irrigation zones.
  - 3) During the performance test, the humidification and irrigation system shall achieve "trouble free" operation of the humidification and irrigation system (i.e. no major breakdown or modification). Operation of all components of the humidification and irrigation system shall also be satisfactory and complete to the specified duty flow and pressure requirements.
  - 4) If during the spray testing runs the humidification and irrigation system does not meet the requirements specified, rectification works shall be undertaken. The initial spray testing shall be considered void and shall be re-run. Complete detailed log of all final tests shall be assembled and submitted to the S.O. on completion of the tests.
- (f) Process Performance Testing
  - (i) General
    - 1) Following completion of Site Acceptance Testing, Process Performance Tests shall be conducted to prove that the equipment performs in accordance with the long-term performance requirements; and
    - 2) The test details, number of test periods and duration of each test period shall be fully in accordance with the Testing and Commissioning Specifications. Deviations are only permitted with the prior express approval of the S.O..
    - 3) All laboratory analyses shall be carried out by a third-party laboratory accredited to ISO/IEC 17025, engaged by the Contractor and subject to S.O.'s approval.

- (ii) Carbon Performance Verification
  - 1) Carbon Testing: All carbon testing shall be done by an accredited laboratory, as approved by the S.O.;
  - 2) Preliminary Performance Verification: The Contractor shall supply a certificate from the manufacturer certifying that his proposed activated carbon will meet the specifications herein. The certificate shall be submitted by the Contractor to the S.O. for review and approval at least four weeks prior to purchase of the activated carbon;
  - 3) Testing of Carbon to be Delivered: Each lot of activated carbon prior to delivery to the job site shall be accompanied by an analysis sheet of the measured characteristics of the carbon in that lot. One random sample per five thousand (5,000) kilograms of the delivered carbon will be selected by the S.O. for testing to ensure product quality; and
  - 4) If a random sample fails to meet the specifications, the Contractor shall have the option of removing the lot of carbon from the Works location within forty-eight (48) hours of notification or testing another random sample at the Contractors sole expense. If the retest fails, the lot of carbon shall be removed within eight (8) hours of notification. Under no circumstances will more than one retest be permitted.
- (iii) Odour Treatment Unit Performance Testing for Contaminant (H<sub>2</sub>S, VOC, and Odour) – Applicable for all treatment units including Chemical Scrubbers, Biotrickling Filters, Biofilters, and Activated Carbon Filters.
  - 1) The Contractor shall include in their scope of work all sampling and laboratory analysis tasks. The Contractor shall supply, install, and maintain (if required) all test materials, temporary test equipment, consumables and experienced personnel required to demonstrate compliance with the specification.
  - 2) The functional performance of the OCS shall be certified (by the Contractor) to be correct before commencing testing of contaminant removal efficiency. The performance tests shall not be carried out until the S.O. is satisfied that the equipment has reached its optimum operational performance. Prior to making any changes to the odour system that could affect equipment or processes external to the OCS, the Contractor shall advise the S.O. of the intended change.
  - 3) A Performance Testing Plan shall be submitted to the S.O. for review and approval at least FOUR weeks prior to commencing any tests. The required test equipment and apparatus, calibration and setup procedures, methods to obtain all measurements and make calculations shall be included in the Performance Test Procedures, and shall be carried out to the satisfaction of the S.O. The Performance Testing Plan shall include the verification of the guaranteed removal of H<sub>2</sub>S, odour, and VOC as defined by:
    - Successful operation of each item of the OCS system;
    - Overall system operation meets the performance guarantees described in the Particular Specifications for the site; and

- Inlet gas contaminant removal performance meets the criteria provided in the Particular Specifications for the site.
- 4) Prior to performing any tests, the Contractor shall arrange with the S.O. to inspect the installation. Once the following prerequisites are completed, the S.O. will confirm a time for conducting the performance tests:
- The Contractor certifies that the Odour Treatment Unit has been installed correctly;
  - The Contractor certifies that airflow balancing and system functional testing of all components of the Odour Control System has been completed satisfactorily;
  - All documentation associated with commissioning have been received and approved by the S.O.; and
  - All test equipment used during tests shall have a current calibration certificate (issued within the preceding six (6) months) verifying its accuracy.
- 5) The S.O. will have personnel present during the testing who may take any independent samples, record procedures and operating conditions to confirm the measurements taken by the Contractor. The S.O. has the right to observe, sample, and carry out any independent tests in parallel to the Contractor during performance testing. The S.O. shall provide full access and cooperation with the Contractor in the undertaking of these tests.
- 6) During the performance test period, the following requirements are to be observed and measurements recorded:
- The treated air quality parameters are achieved;
  - The gas flow rate and the odour scrubber inlet gas pressure shall be maintained as close as practical (+/- 5%) to the design conditions. The Contractor must be satisfied that the air flow rate through the odour scrubber system is in accordance with the design gas flow rates and conditions stipulated in the Particular Specifications before commencing the performance tests;
  - The gas flow rate and the pressure loss across the odour scrubber vessels shall be either recorded continuously by the online measurement (if applicable), or measured and recorded manually, twice per day, e.g. once in the morning and once in the afternoon;
  - The inlet and outlet gas temperature and humidity are to be either recorded continuously by the online measurement (if applicable), or measured and recorded twice per day, e.g. once in the morning and once in the afternoon;
  - Other operating parameters such as pH, conductivity, any unusual events, etc., are to be recorded for each day of the test; and

- No manual adjustment of valves or instrumentation associated with the scrubber vessels will be allowed during the test period.
- 7) Following analysis of the H<sub>2</sub>S data to determine the days and times of sample collection, five (5) air samples shall be taken on the inlet and outlet of the odour control system for three (3) consecutive days (total of 30 samples) at greater than one-hour intervals to reflect the peak odour loads within the performance test period, as approved by the S.O. A minimum of thirteen (13) out of the fifteen (15) outlet readings shall meet the performance criteria as specified and the average of all the daily outlet results shall meet the concentration and removal performance criteria as specified in Annex O – Schedule of Performance Guarantees.
- 8) The collection and analysis of the air samples shall be carried out in accordance with the requirements of EN 13725.
- 9) The Contractor shall prepare and maintain the performance test data log sheets with a photographic record of the sampling technique and sampling location for each collected sample. The sampling log sheets shall be maintained that will include the following as a minimum:
- Date, time, location, sampler, and results of each sample.
  - A description of each photograph taken.
  - Weather conditions for the sampling day.
  - A qualitative description of plant operations.
  - A description of any deviations from the sampling plan.
- 10) Unless otherwise specified in the Particular Specifications, within TEN working days of the completion of the test period, the Contractor shall provide a written test report showing all laboratory and field test results. The report shall include the following as a minimum:
- Narrative of the testing activities.
  - All laboratory and field test data including on-line instrumentation trends, a copy of the original sampling log, photographs showing locations of flow and pressure measurements, tabular summary of velocity, airflow rates, pressures, H<sub>2</sub>S and odor removal data, and calculated results.
  - Description of all deviations to the Performance Testing Plan.
  - The conclusions of the tests with reference to the performance criteria.
- 11) At the completion of the performance tests copies of the performance test data log sheets shall be provided by the Contractor to the S.O..

(iv) Specific Requirements for Performance Testing Sample Collection and Analysis

1) H<sub>2</sub>S Sampling and Testing Requirements

- Continuous H<sub>2</sub>S monitoring instruments for the odour scrubber system inlet and outlet gas streams will be provided by the Contractor during the test period. The inlet and outlet readings will be recorded for the OCF performance test period described in G4.72.16(f)(iii) and be adequate to demonstrate compliance with the Schedule of Process Guarantees);
- H<sub>2</sub>S data logger units of a suitable sensitivity range shall be provided by the Contractor and installed at the inlet and outlet ducts/plenums of each odour scrubber stage to allow verification of the H<sub>2</sub>S removal efficiency of the Biofilters/Biotrickling Filters/Chemical Scrubbers or the AC system individually. Ideally the location of the H<sub>2</sub>S data logger units shall ensure representative samples of the combined odour treatment system and/or each individual odour treatment stage. If this is not possible then at least TWO individual units selected by the S.O. shall be used. Each location shall have at least TWO H<sub>2</sub>S data logger units installed;
- All H<sub>2</sub>S data logger units used for this exercise shall have proof of calibration undertaken within the last six (6) months. Proof of calibration shall be submitted to the S.O. prior to their installation;
- H<sub>2</sub>S data logged by the data logger units shall be logged in one (1) minute intervals;
- All H<sub>2</sub>S data logger units, their software, and calibration equipment shall remain the property of the Contractor at the completion of the proof of performance tests;
- At the end of the logging period, the Contractor shall check the quality and continuity of the data and submit the raw data to the S.O. for analysis.
- Upon review of the H<sub>2</sub>S logging data, the S.O. will advise the Contractor of times and days that air sampling shall be undertaken for odour, VOC, and RSC;
- H<sub>2</sub>S analysis using a handheld H<sub>2</sub>S Analyser of adequate sensitivity (to 50 ppb) shall also be undertaken in the laboratory for all collected odour samples.
- All H<sub>2</sub>S data logger units shall be reinstalled at their original location following completion of data downloading such that data collection can continue until after the odour testing is completed;

2) VOC Sampling and Testing Requirements

- The number of samples to be taken, collection interval and test period shall be as specified in G4.72.16(f)(iii) as described above, and be adequate to demonstrate compliance with the Schedule of Process Guarantees in the Particular Specifications;

- During sample collection, the inlet and outlet samples should be taken at approximately the same time to allow removal efficiency performance to be calculated and each set of inlet/outlet samples shall be taken at greater than ONE (1) hour intervals to reflect the peak VOC load; and
  - The inlet sample and one outlet sample are to be subjected to a Gas Chromatography Mass Spectrometry (GCMS) analysis for VOCs. The samples shall be analysed within 24 hours of collection, and samples shall be kept out of direct sunlight. Timing of the sampling shall be agreed between the Contractor and the S.O..
- 3) Reduced Sulphur Compounds (RSC) Sampling and Testing Requirements
- The number of samples to be taken, collection interval and test period shall be as specified in G4.72.16(f)(iii) as described above, and be adequate to demonstrate compliance with the Schedule of Process Guarantees outlined in the Particular Specifications;
  - During sample collection, the inlet and outlet samples should be taken at approximately the same time to allow removal efficiency performance to be calculated and each set of inlet/outlet samples shall be taken at greater than ONE (1) hour intervals to reflect the peak RSC load; and
  - The inlet sample and one outlet sample are to be subjected to a Gas Chromatography coupled with Chemiluminescence Detection (GC-SCD) or a Flame Ionisation Detector (FID). The samples shall be analysed within 24 hours of collection, and samples shall be kept out of direct sunlight. Timing of the sampling shall be agreed between the Contractor and the S.O.
- 4) Odour Sampling and Testing
- The number of samples to be taken, collection interval and test period shall be as specified in G4.72.16(f)(iii) as described above, and be adequate to demonstrate compliance with the Schedule of Process Guarantees outlined in the Particular Specifications;
  - During sample collection, the inlet and outlet samples should be taken at approximately the same time to allow removal efficiency performance to be calculated. Each set of inlet/outlet samples shall be taken at greater than ONE (1) hour intervals to reflect the peak odour load;
  - Each odour sample shall be measured both on site during sampling and in laboratory using a handheld H<sub>2</sub>S analyser of adequate sensitivity (to 50 ppb);

- Care shall be taken to prevent the contamination of the sampling and analysis equipment of the low odour samples by the high odour samples. The order of odour sample collection shall start with the sources expected to have the lowest odour concentrations, and progressively move onto the sources with the higher odour concentrations; and
- Odour samples shall be analysed by a qualified odour laboratory to CEN-EN13725 and be analysed for odour concentration, intensity, and hedonic tone.
- The odour samples shall be analysed within 24 hours of collection and shall not be exposed to direct sunlight before analysis.

5) Requirements for Results Analysis

- Odour removal performance for each of the inlet/outlet sample sets during test period shall be calculated using each individual odour concentration as determined by a 3<sup>rd</sup> party lab accredited to ISO/IEC 17025, such as the Singapore Accreditation Council (SAC) Singapore Laboratory Accreditation Scheme (SINGLAS) laboratories;
- H<sub>2</sub>S removal performance shall be calculated from the H<sub>2</sub>S data logger monitoring results and verified from the H<sub>2</sub>S sampling results recorded in the laboratory;
- VOC removal performance shall be calculated from the VOC GCMS sampling results recorded in the laboratory and verified from the SCADA logging results (if available);
- RSC removal performance shall be calculated from the RSC-GC-SCD or GC-FID sampling results recorded in the laboratory;

6) Pass/Fail Criteria

- The performance tests shall be successful if, for each day of stipulated test period:
  - Sample outlet odour, H<sub>2</sub>S, VOC and RSC concentrations do not exceed the values specified in the Schedule of Performance Guarantees stipulated in the Particular Specifications;
  - The daily averages of all the calculated pollutant removal efficiencies as recorded from the sampling results of the OCS for H<sub>2</sub>S, odour, VOC, and RSC are greater than the removal performance criteria as specified in the Schedule of Performance Guarantees stipulated in the Particular Specifications;
  - The calculated contaminant removal efficiency for the OCS for H<sub>2</sub>S, odour, VOC, and RSC from the sampling results are greater than the values specified in the Schedule of Performance Guarantees stipulated in the Particular Specifications, for at least 80 % of all collected samples;

- In addition to the above, for parameters at locations with the ability to record continuous online concentrations at both the inlet and outlet ducts such as H<sub>2</sub>S, the following shall apply in defining the success of Performance Tests:
  - All available continuous online H<sub>2</sub>S concentrations at the OCS inlet and outlet (at stack or Biofilter outlet) as recorded by the PLC and SCADA shall be recorded and analysed to determine the hourly average OCS outlet concentrations and hourly average contaminant removal rates.
  - The calculated hourly average OCS outlet H<sub>2</sub>S concentrations shall not exceed the values specified in Annex O – Schedule of Performance Guarantees and stipulated in the Particular Specifications for more than five percent (5%) of the time. No more than two (2) exceedances are allowed per day for each individual hourly average OCS outlet concentration data.
  - For the remaining five (5) percent where the calculated hourly average OCS outlet concentration H<sub>2</sub>S concentration exceeds the values specified in Annex O – Schedule of Performance Guarantees and stipulated in the Particular Specifications, the calculated hourly average contaminant removal efficiencies must not exceed the values specified in Annex O – Schedule of Performance Guarantees and stipulated in the Particular Specifications.
  - The calculated hourly average contaminant removal efficiencies shall not exceed the values specified in Annex O – Schedule of Performance Guarantees and stipulated in the Particular Specifications at any time the OCS inlet H<sub>2</sub>S concentration falls within the expected inlet gas concentration range outlined in the Particular Specifications.

**END OF SECTION**